

Uzbekistan Regional GHG Emissions 2022

Spatial Disaggregation — Methodology Comparison Report

v1: ESRI 10m LULC + VIIRS Nightlights | v2: Sentinel-5P Atmospheric Signals + LULC + VIIRS (reliability-adjusted)

v1 — LULC + Nightlights

Spatial proxy signals:

- * ESRI Global LULC 10m Time Series (200m mosaic)
- * NOAA/VIIRS monthly nightlights, annual median 2022

Disaggregation logic:

Each sector allocated by LULC class fractions
(built area, crops, bare ground ...) weighted by
nightlight intensity as an economic activity proxy.

Strengths:

- + Deterministic and reproducible
- + Not affected by atmospheric retrieval noise
- + Excellent sub-city spatial resolution (200m)

Limitations:

- Cannot distinguish emission intensity within the same LULC class (e.g. heavy vs light industry)
- Nightlights proxy economic activity, not chemistry
- No sector-specific atmospheric signal

v2 — Sentinel-5P + LULC + Nightlights (reliability-adjusted)

Spatial proxy signals (TROPOMI OFFL L3 annual medians):

- * N02 tropospheric column (cloud_fraction < 0.5)
- * S02 total column
- * CO total column
- * CH4 dry-air mixing ratio, bias-corrected (anomaly above P10)
- * AER_AI absorbing aerosol index
- + ESRI LULC 10m + VIIRS nightlights (same as v1)

Per-sector S5P blend weights (nominal):

- Elec & Heat : S02 40% + N02 25% + built 15% + NL 20%
- Transport : N02 45% + CO 35% + built 20%
- Fugitive : CH4 55% + S02 25% + uniform 20%
- Agriculture : CH4 40% + crops 40% + AER 20%

Regional reliability adjustment:

r = clamp(1 - sum(noise_weight * LULC_frac), 0.15, 1.0)
S5P weight *= r; deficit redistributed to LULC/NL.

Strengths:

- + Actual atmospheric chemistry as primary constraint
- + Sector-specific signal (S02=power, N02=traffic ...)

Limitations:

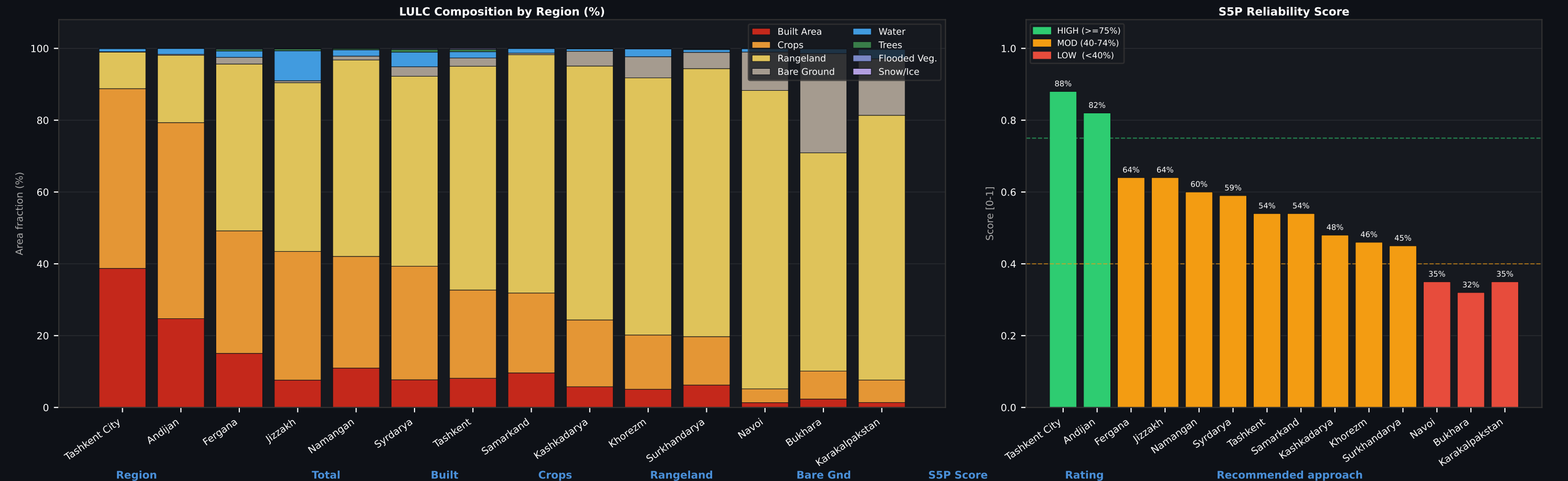
- TROPOMI ~5.5 km footprint limits sub-city detail
- Desert / bright surfaces degrade UV/VIS retrievals

Report structure: Page 1 Cover | Page 2 Summary Table | Pages 3-16 Per-region comparison | Page 17 Analytics | Page 18 Findings

Emission totals: Uzbekistan First Biennial Transparency Report (BTR1) 2024 | LULC: ESRI Global LULC 10m | NL: NOAA/VIIRS DNB | S5P: Copernicus OFFL L3 TROPOMI

Scientific basis for reliability scoring: van Geffen et al. 2022 (NO2 ATBD v2.4); Theys et al. 2017 ACP (SO2); Torres et al. 2007 JGR (AER_AI); Tilstra et al. 2021 AMT (surface LER climatology)

Regional Overview — LULC Composition & S5P Reliability

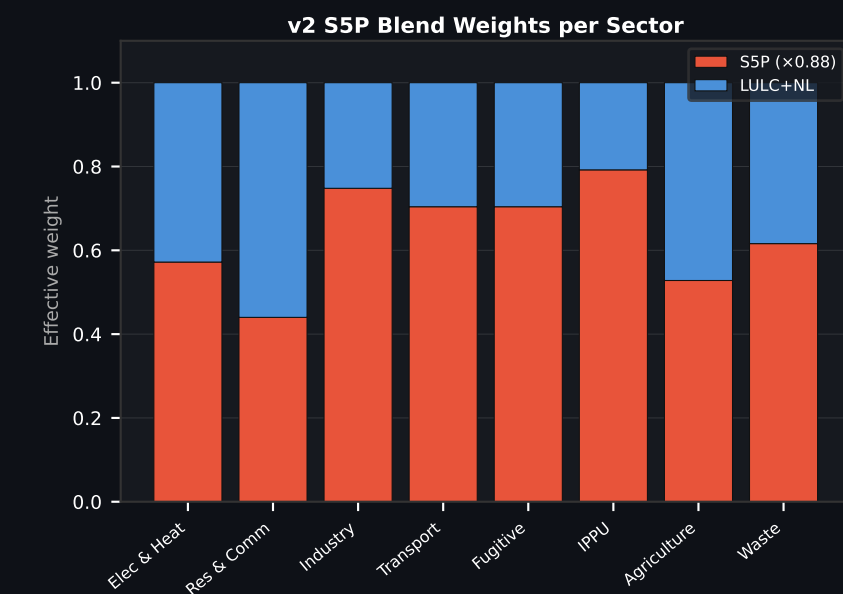
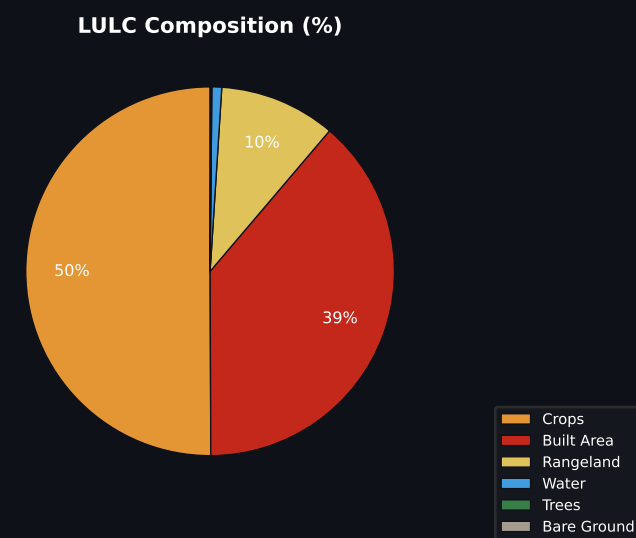
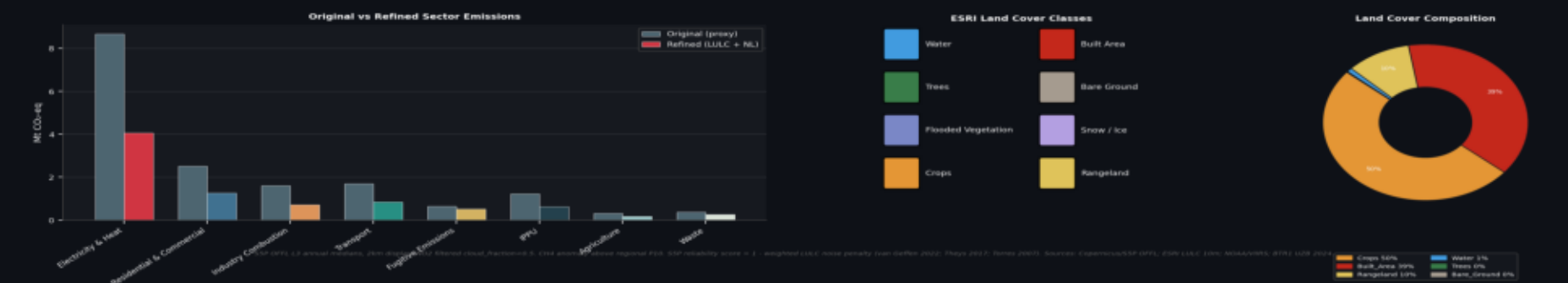
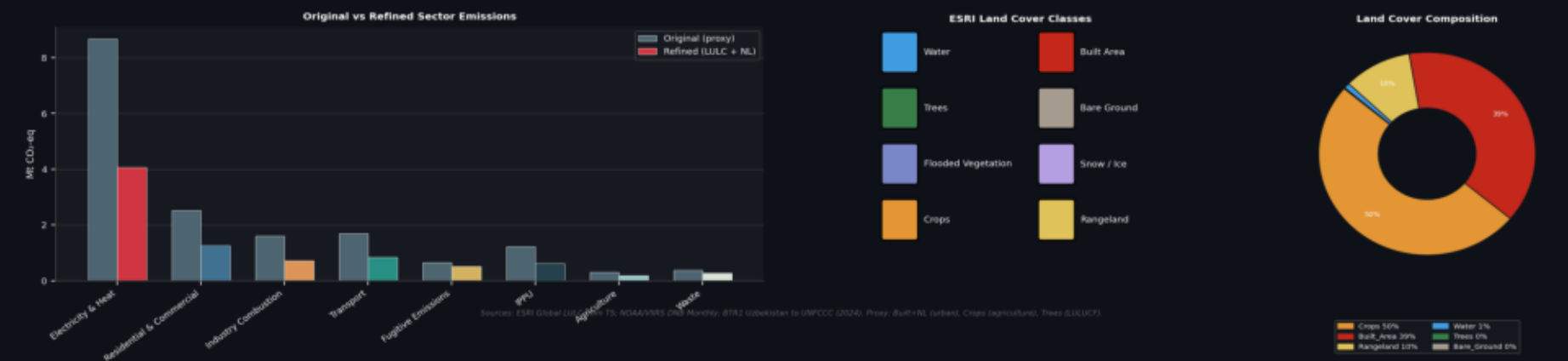
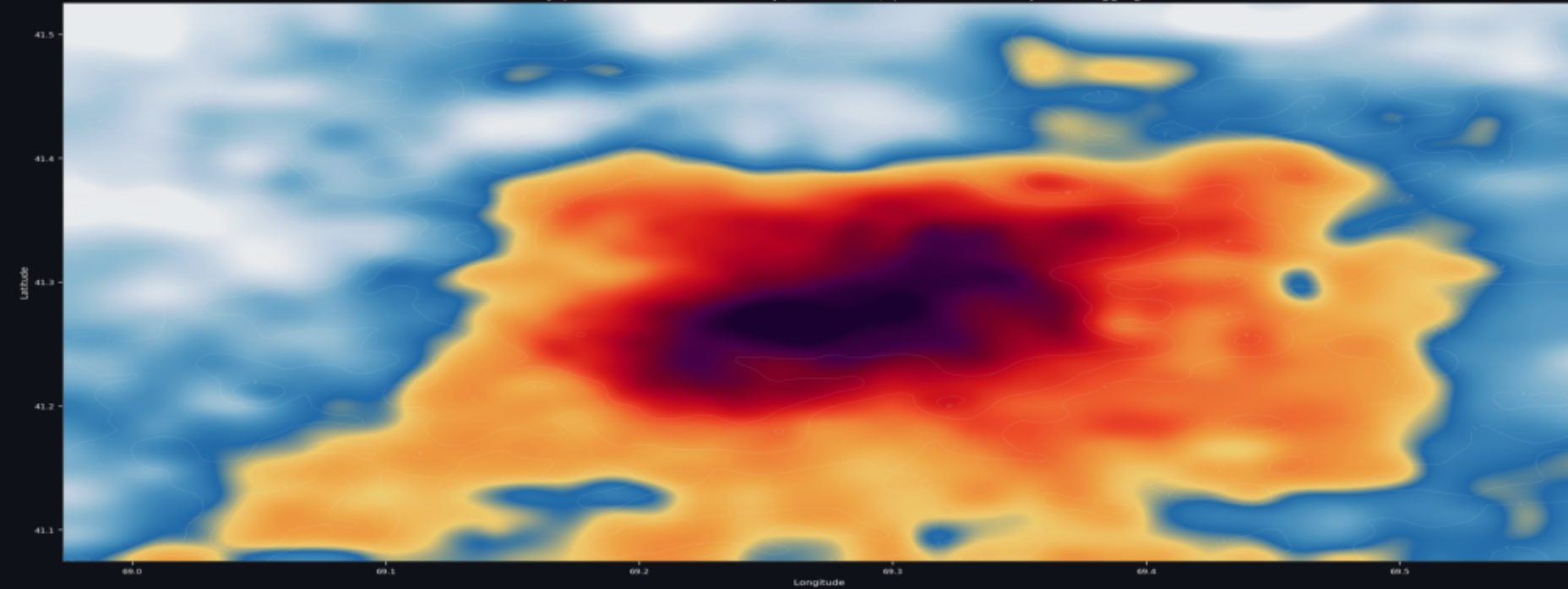


Region	Total (Mt CO2-eq)	Built (%)	Crops (%)	Rangeland (%)	Bare Gnd (%)	S5P Score	Rating	Recommended approach
Tashkent City	17.03	38.7	50.1	10.2	0.1	0.88	HIGH	v2 S5P primary
Andijan Region	12.65	24.7	54.6	18.7	0.3	0.82	HIGH	v2 S5P primary
Fergana Region	12.96	15.1	34.1	46.5	1.9	0.64	MODERATE	v2 blended
Jizzakh Region	6.89	7.6	35.8	47.0	0.5	0.64	MODERATE	v2 blended
Namangan Region	10.29	11.0	31.1	54.7	1.1	0.60	MODERATE	v2 blended
Syrdarya Region	18.39	7.7	31.6	52.9	2.6	0.59	MODERATE	v2 blended
Tashkent Region	17.03	8.1	24.6	62.3	2.3	0.54	MODERATE	v2 blended
Samarkand Region	12.42	9.6	22.2	66.4	0.4	0.54	MODERATE	v2 blended
Kashkadarya Region	32.75	5.8	18.6	70.7	4.1	0.48	MODERATE	v2 blended
Khorezm Region	5.97	5.1	15.1	71.6	5.9	0.46	MODERATE	v2 blended
Surkhandarya Region	11.44	6.2	13.5	74.6	4.5	0.45	MODERATE	v2 blended
Navoi Region	21.39	1.3	3.8	83.1	10.7	0.35	LOW	v1 preferred
Bukhara Region	17.02	2.3	7.8	60.8	27.7	0.32	LOW	v1 preferred
Rep. Karakalpakstan	11.06	1.4	6.3	73.7	15.7	0.35	LOW	v1 preferred

Tashkent City | S5P Reliability: HIGH (88%)

GHG Emissions — Tashkent City (2022) — Sentinel-5P Refined

Tashkent City | Refined total: 8.46 Mt CO₂-eq (excl. LULUCF) | SSP-constrained spatial disaggregation

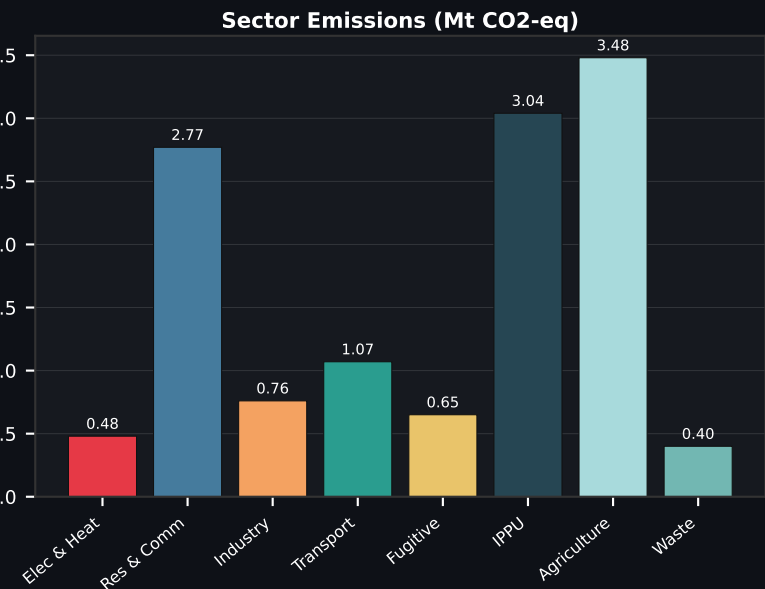
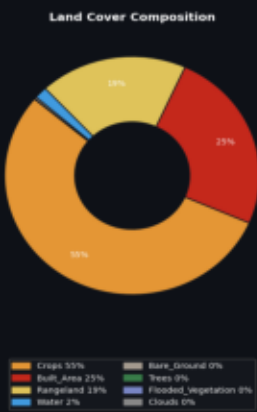
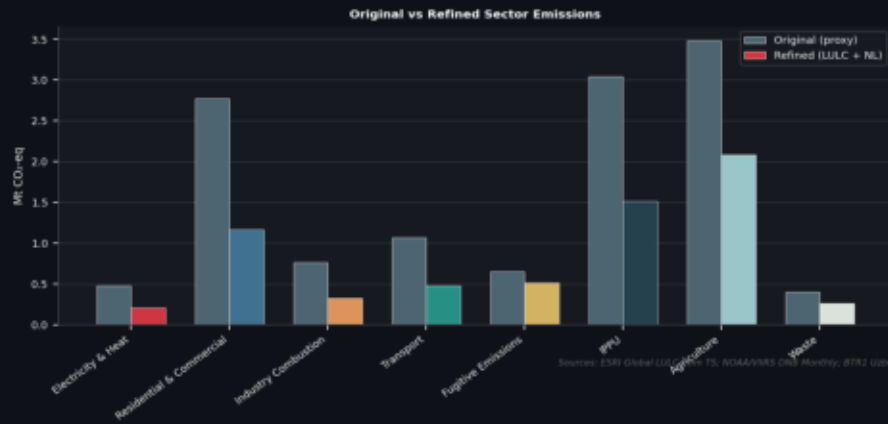
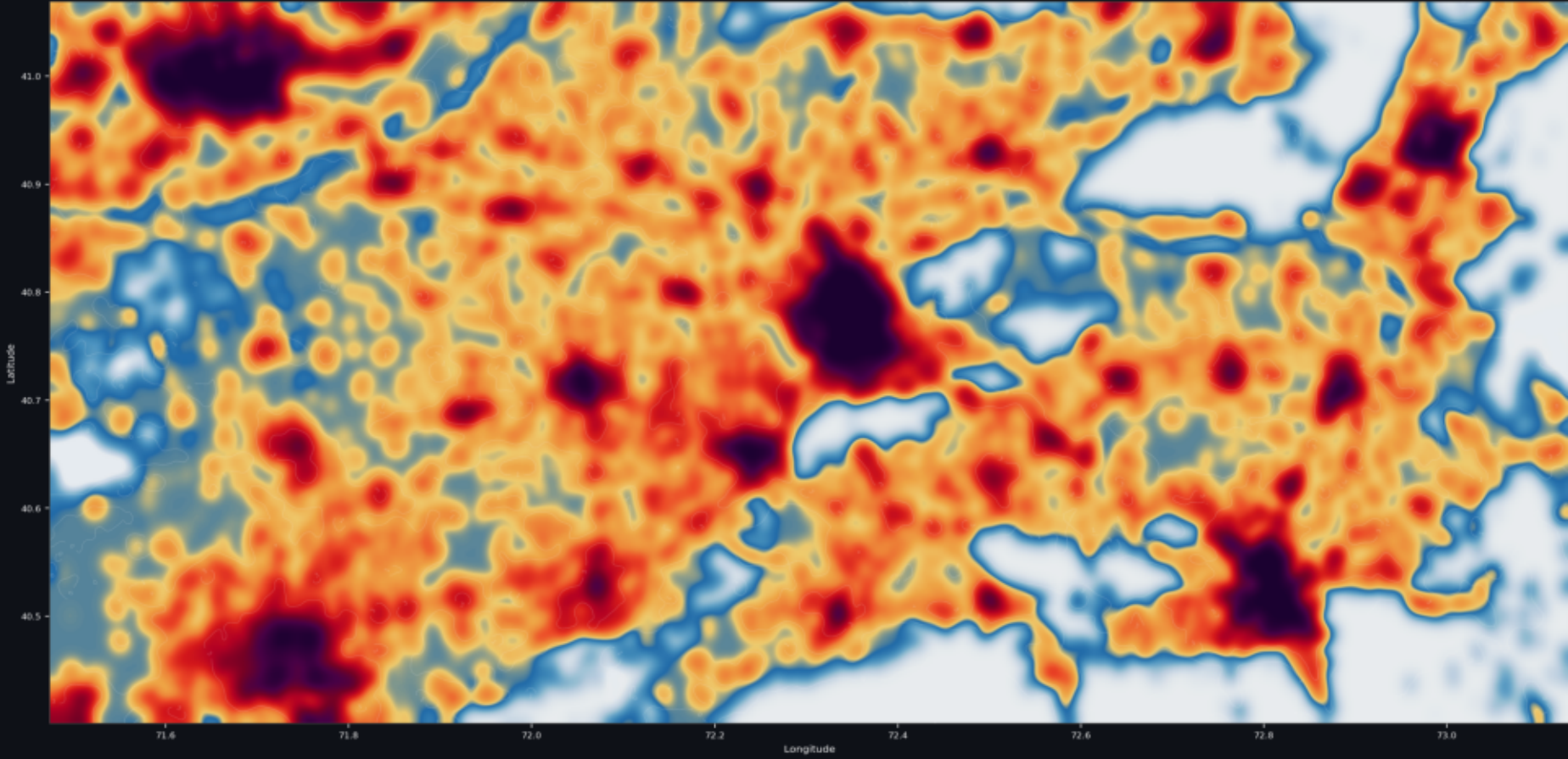


Crops

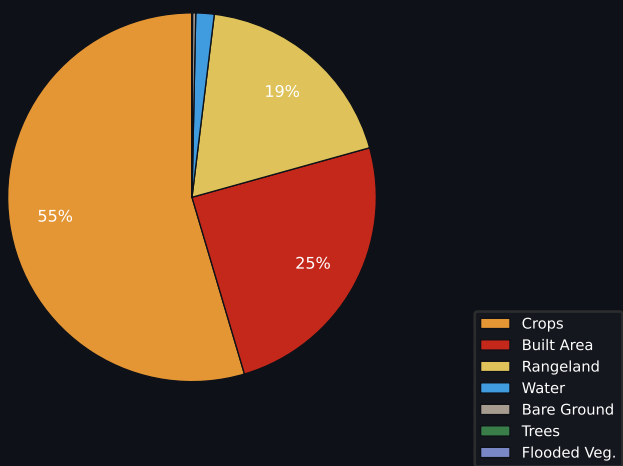
v1 — LULC + Nightlights

GHG Emissions — Andijan Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR1 Uzbekistan proxy-based estimates
Andijan Region — Refined total: 6.54 Mt CO₂-eq (excl. LULUCF)



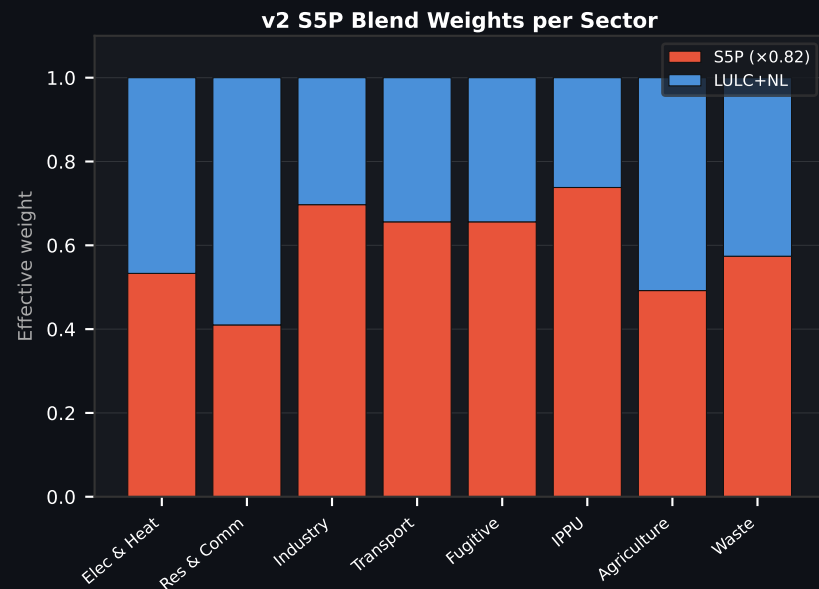
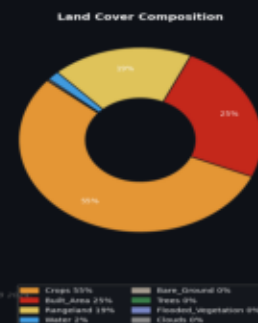
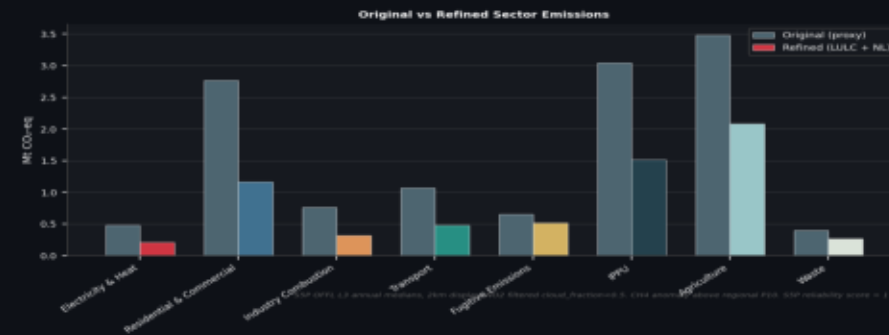
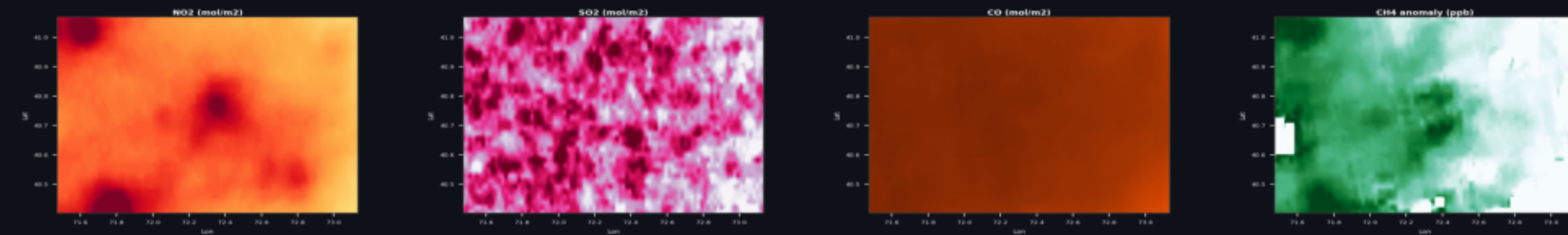
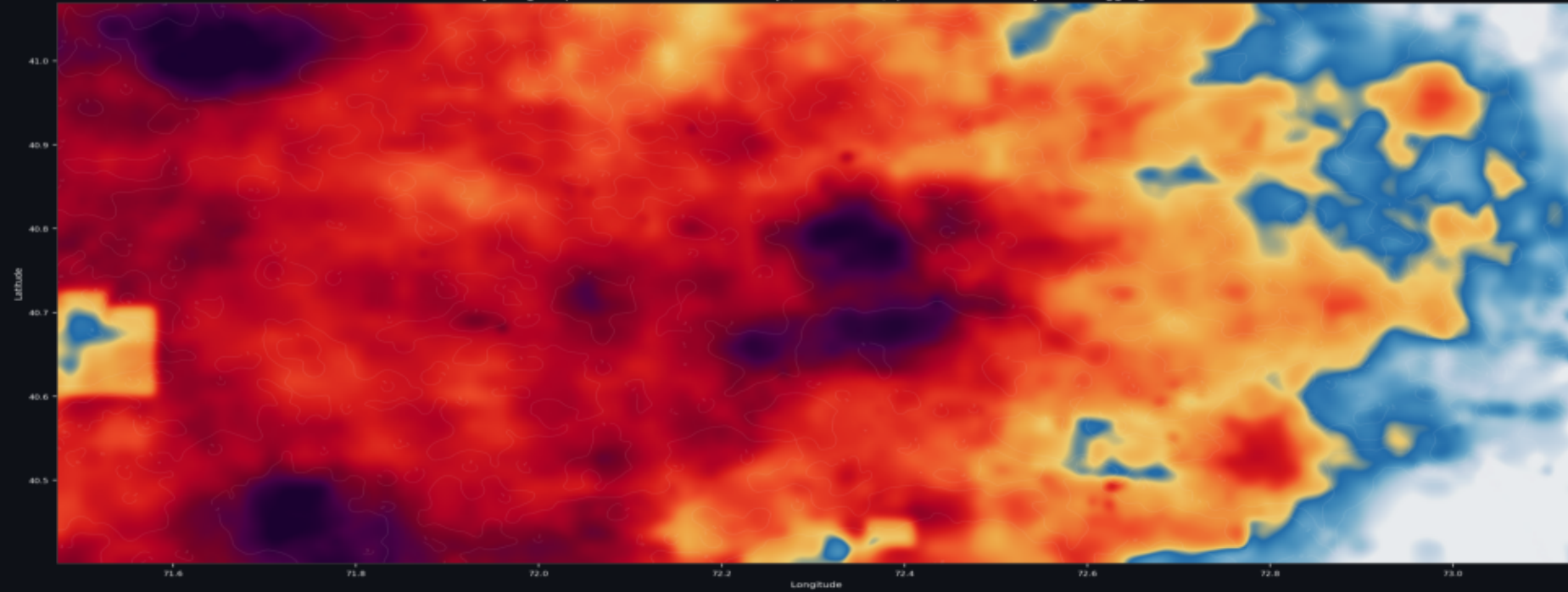
LULC Composition (%)



v2 — S5P + LULC + NL

GHG Emissions — Andijan Region (2022) — Sentinel-5P Refined

S5P weight: 82% of nominal (LULC-adjusted) | Spatial constraint: S5P NO₂/SO₂/CO/CH₄ + ESRI 10m LULC + VIIRS NL | Totals: BTR1 Uzbekistan 2024
Andijan Region | Refined total: 6.54 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation



S5P Reliability:
HIGH (82%)

Approach:
S5P primary

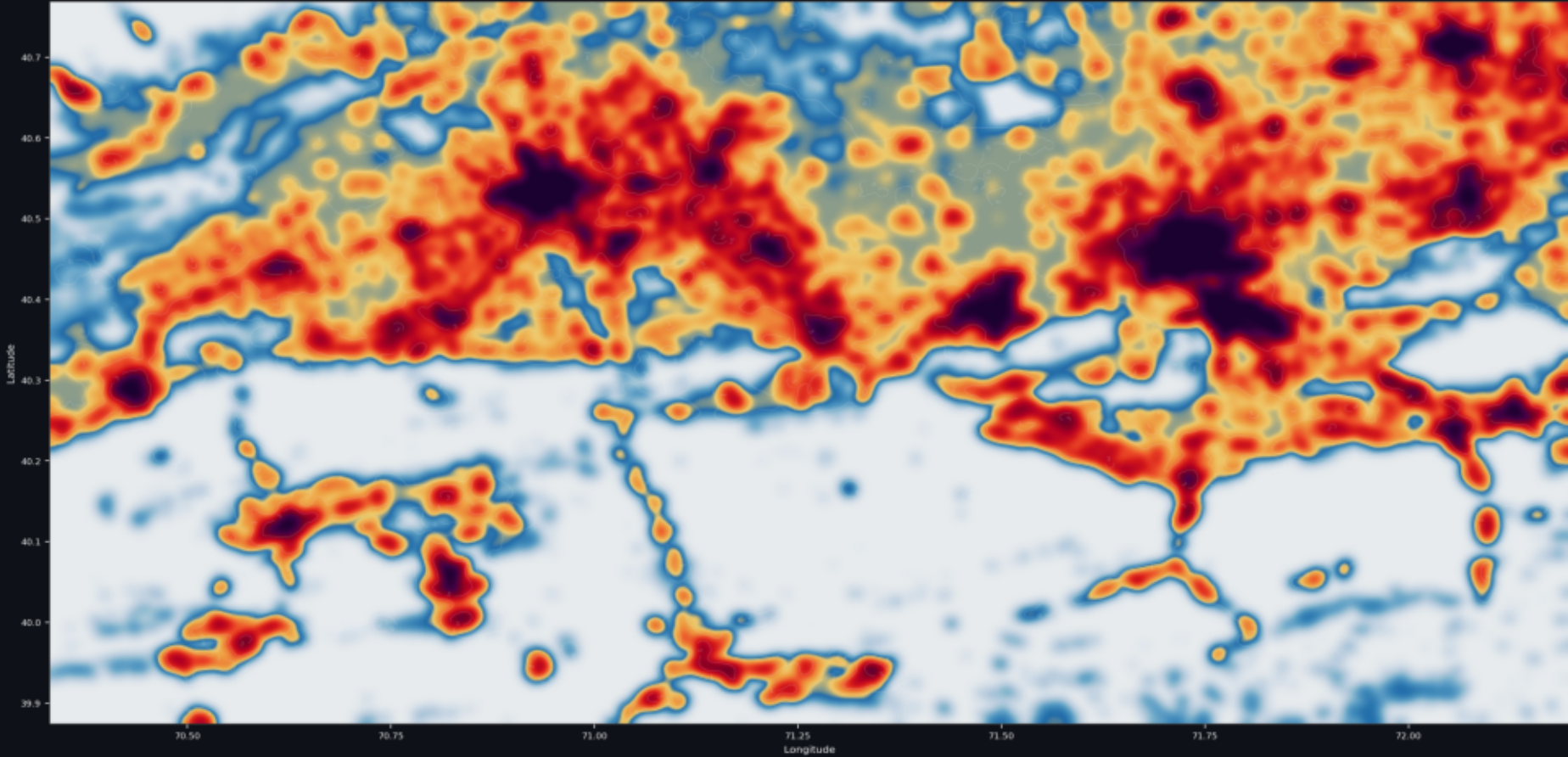
Total emissions:
12.65 Mt CO₂-eq

Dominant sector:
Agriculture

Dominant LULC:
Crops

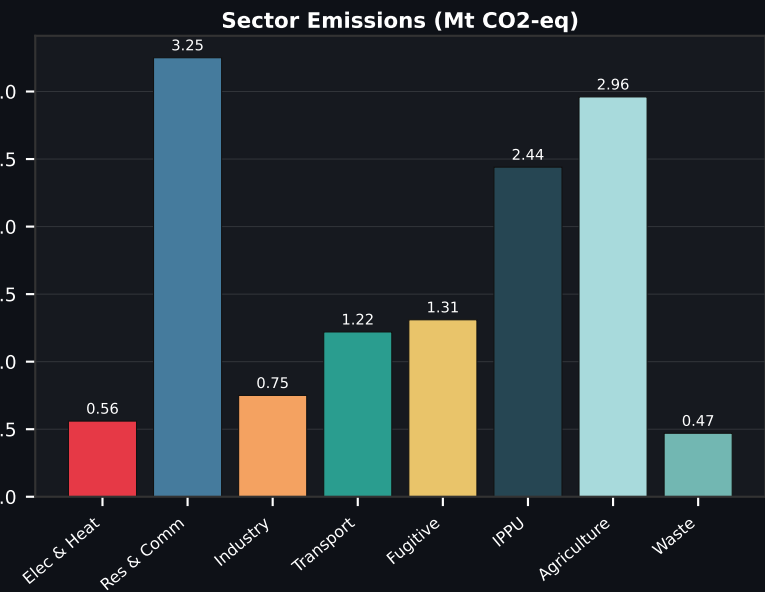
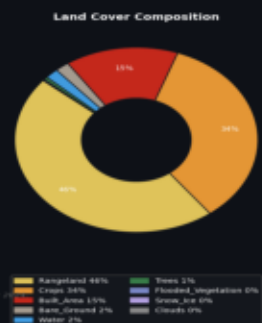
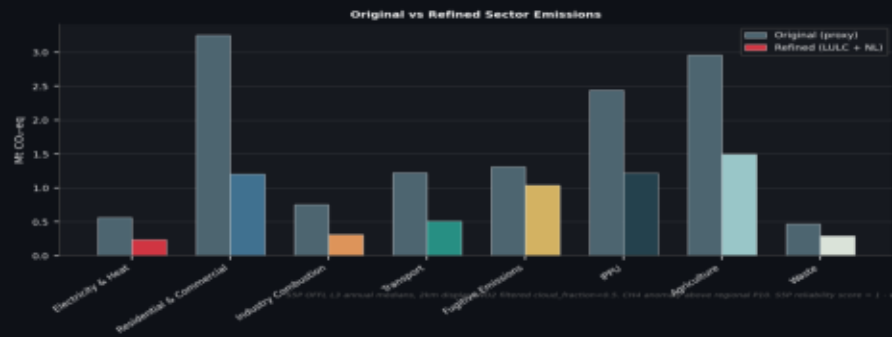
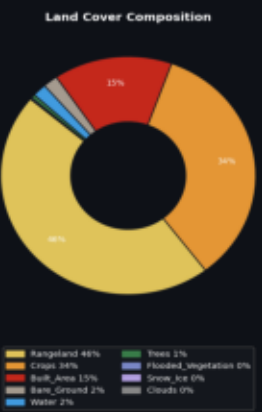
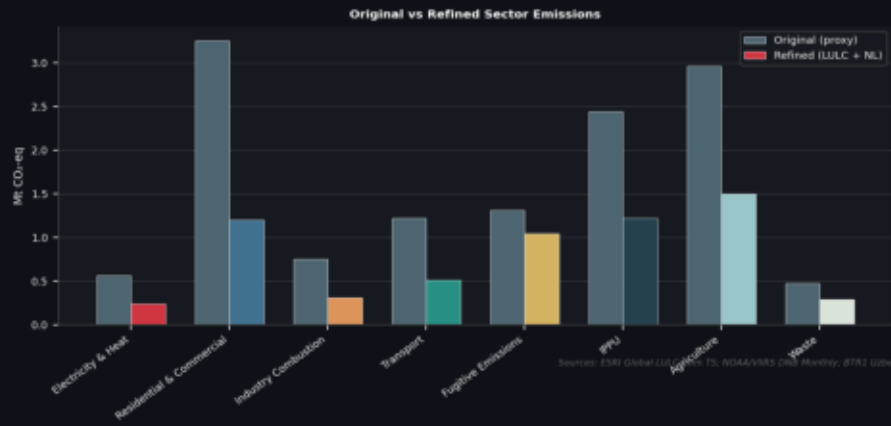
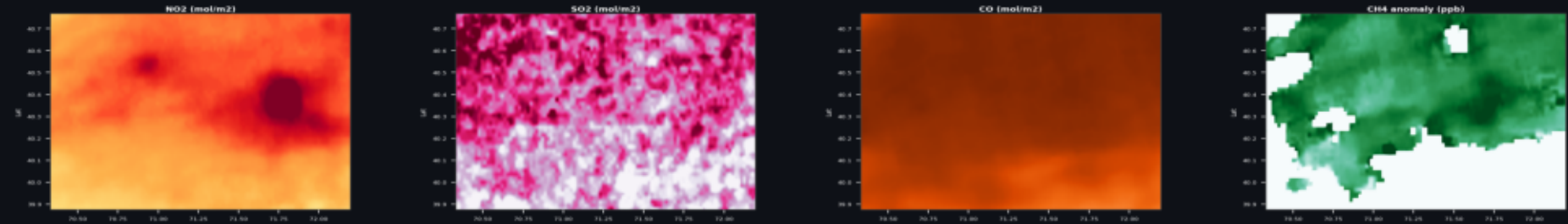
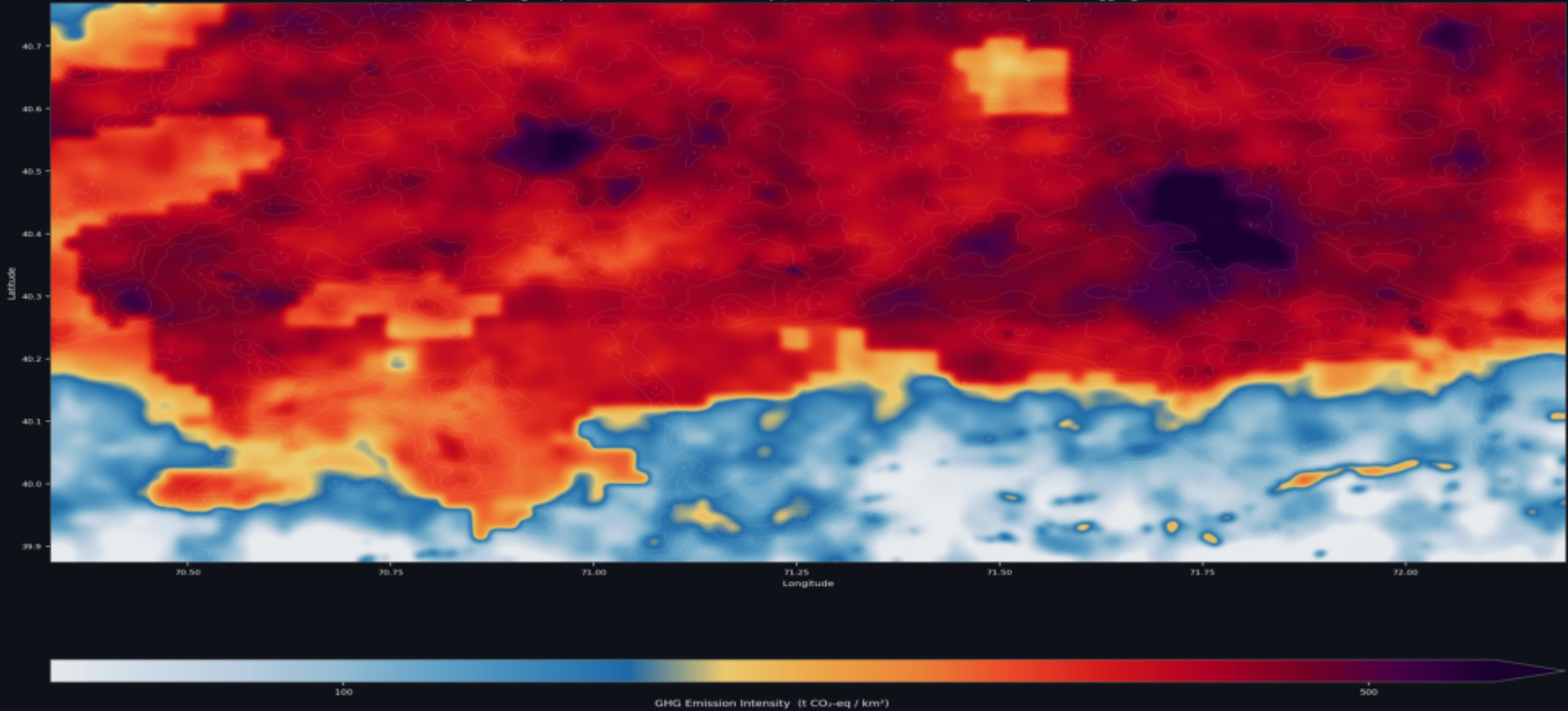
GHG Emissions — Fergana Region (2022)

Spatially disaggregated using ESAI 30m LULC + VIIRS Nightlights | Refined from BTR3 Uzbekistan proxy-based estimates
Fergana Region — Refined total: 6.30 Mt CO₂-eq (excl. LULUCF)

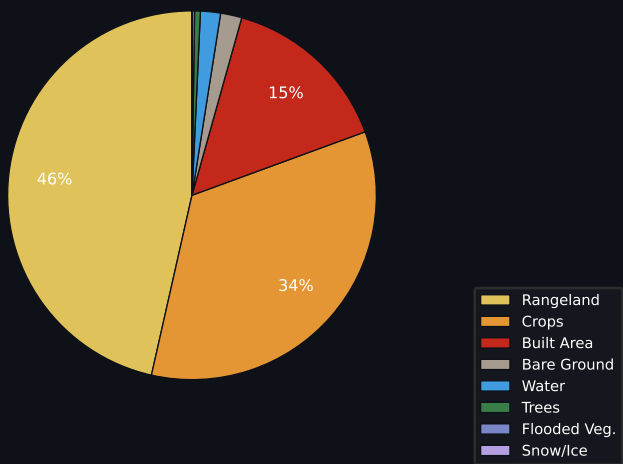


GHG Emissions — Fergana Region (2022) — Sentinel-5P Refined

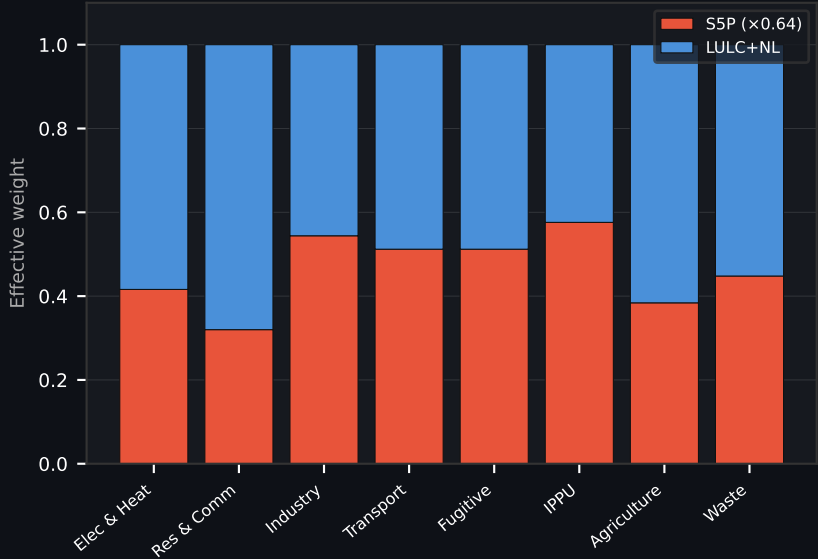
S5P weight: 64% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESAI 10m LULC + VIIRS NL | Totals: BTR3 Uzbekistan 2024
Fergana Region | Refined total: 6.30 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation



LULC Composition (%)



v2 S5P Blend Weights per Sector



S5P Reliability:

MODERATE (64%)

Approach:

Blended

Total emissions:

12.96 Mt CO₂-eq

Dominant sector:

Residential Commercial

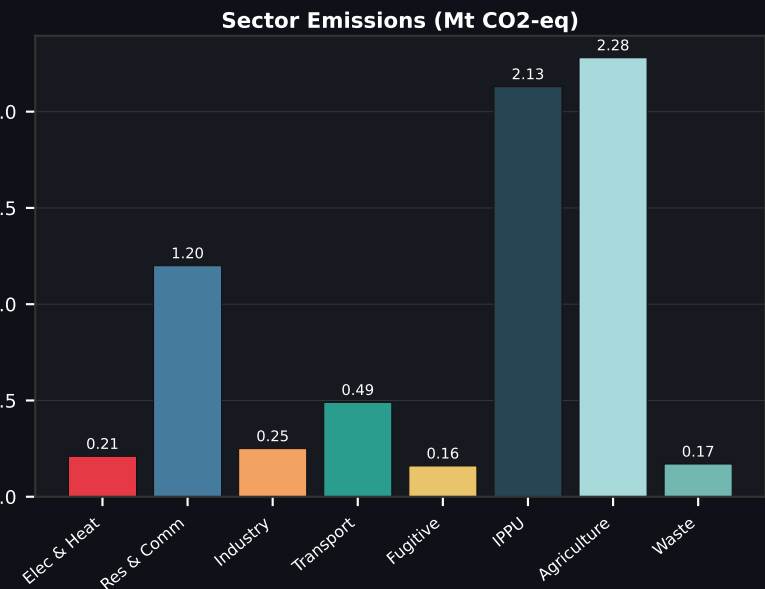
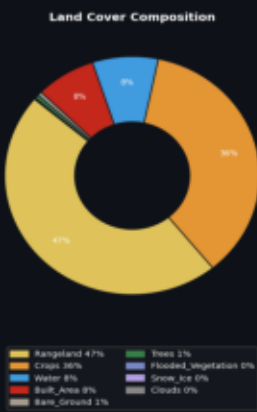
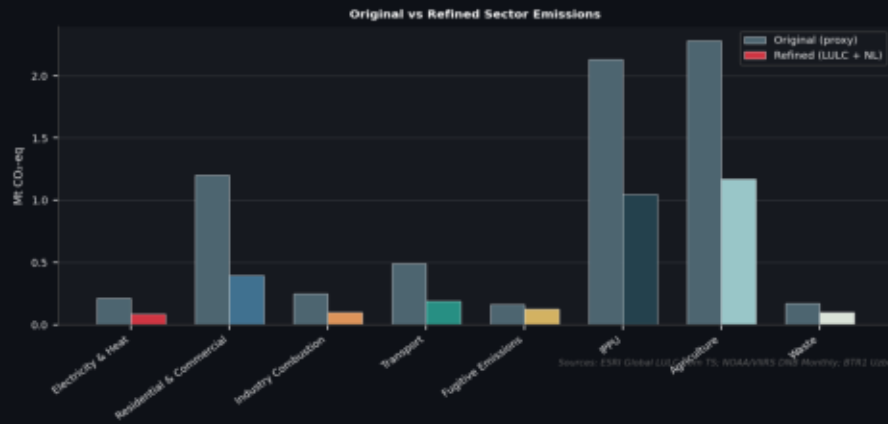
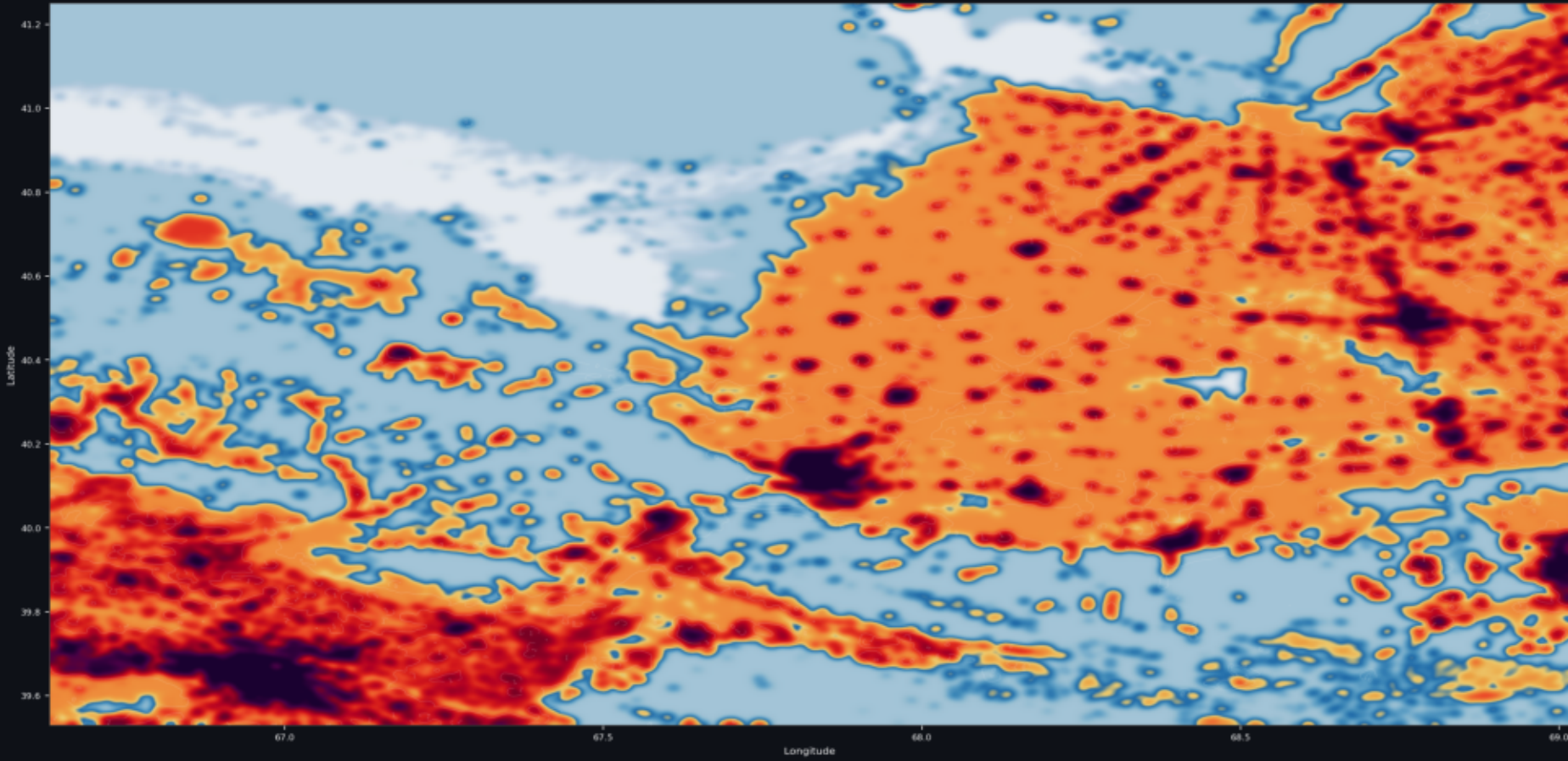
Dominant LULC:

Rangeland

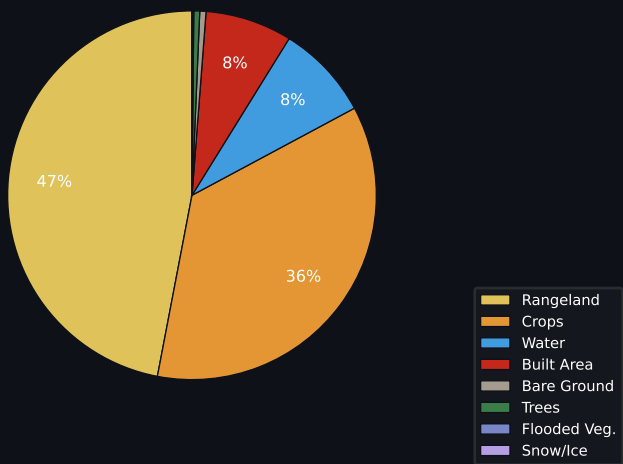
v1 — LULC + Nightlights

GHG Emissions — Jizzakh Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR1 Uzbekistan proxy-based estimates
Jizzakh Region — Refined total: 3.21 Mt CO₂-eq (excl. LULUCF)



LULC Composition (%)

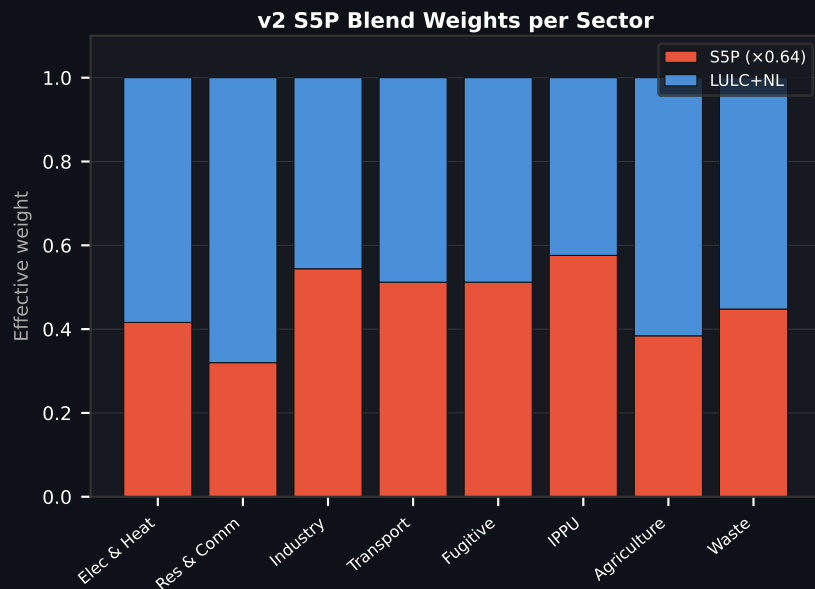
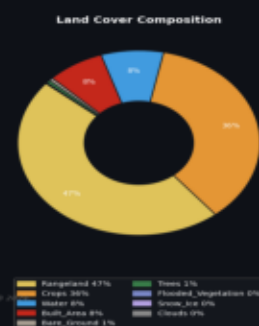
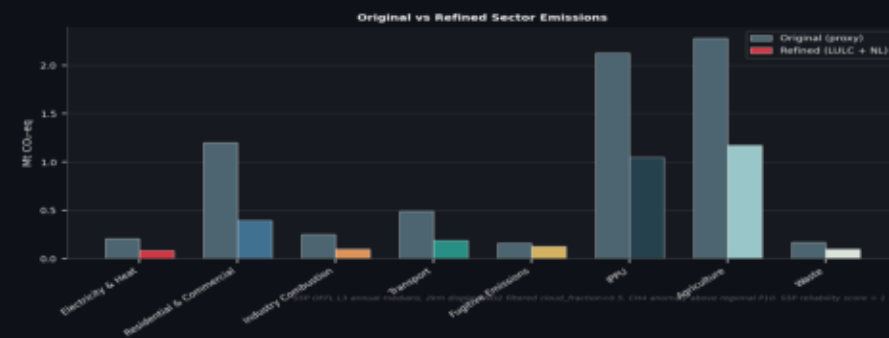
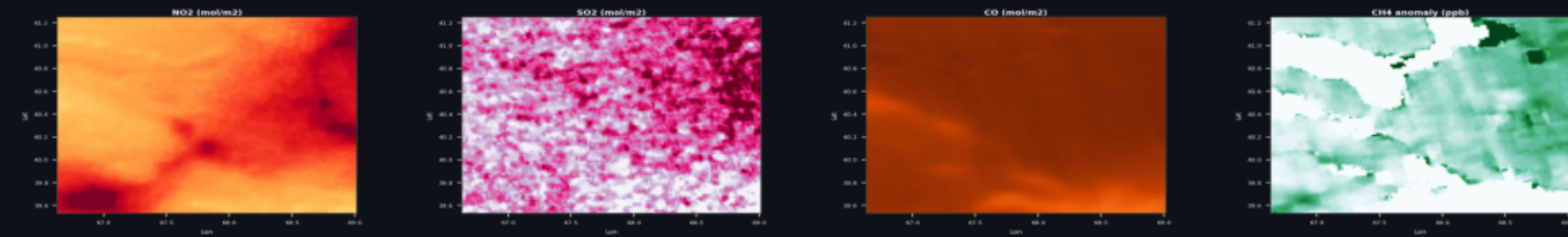
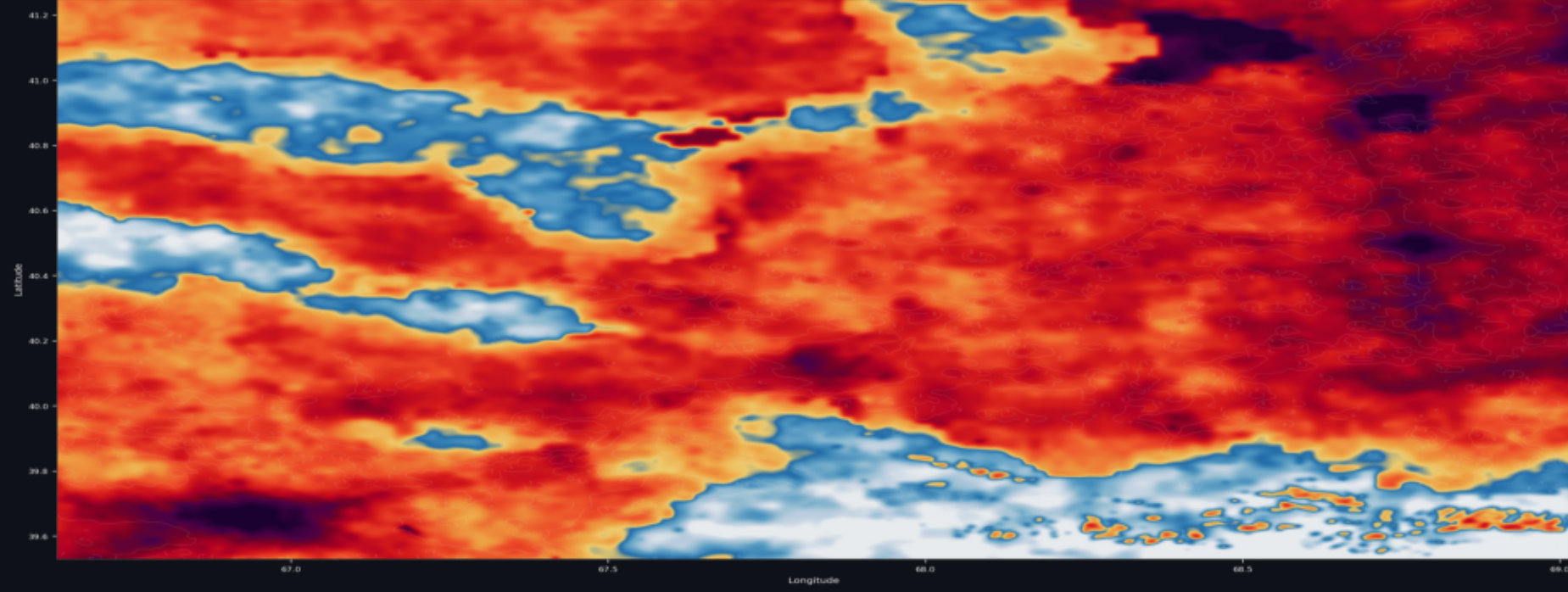


v2 — S5P + LULC + NL

GHG Emissions — Jizzakh Region (2022) — Sentinel-5P Refined

S5P weight: 64% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESRI 10m LULC + VIIRS NL | Total: BTR1 Uzbekistan 2024
Jizzakh Region | Refined total: 3.21 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation

S5P reliability: MODERATE (64%)



S5P Reliability:

MODERATE (64%)

Approach:

Blended

Total emissions:

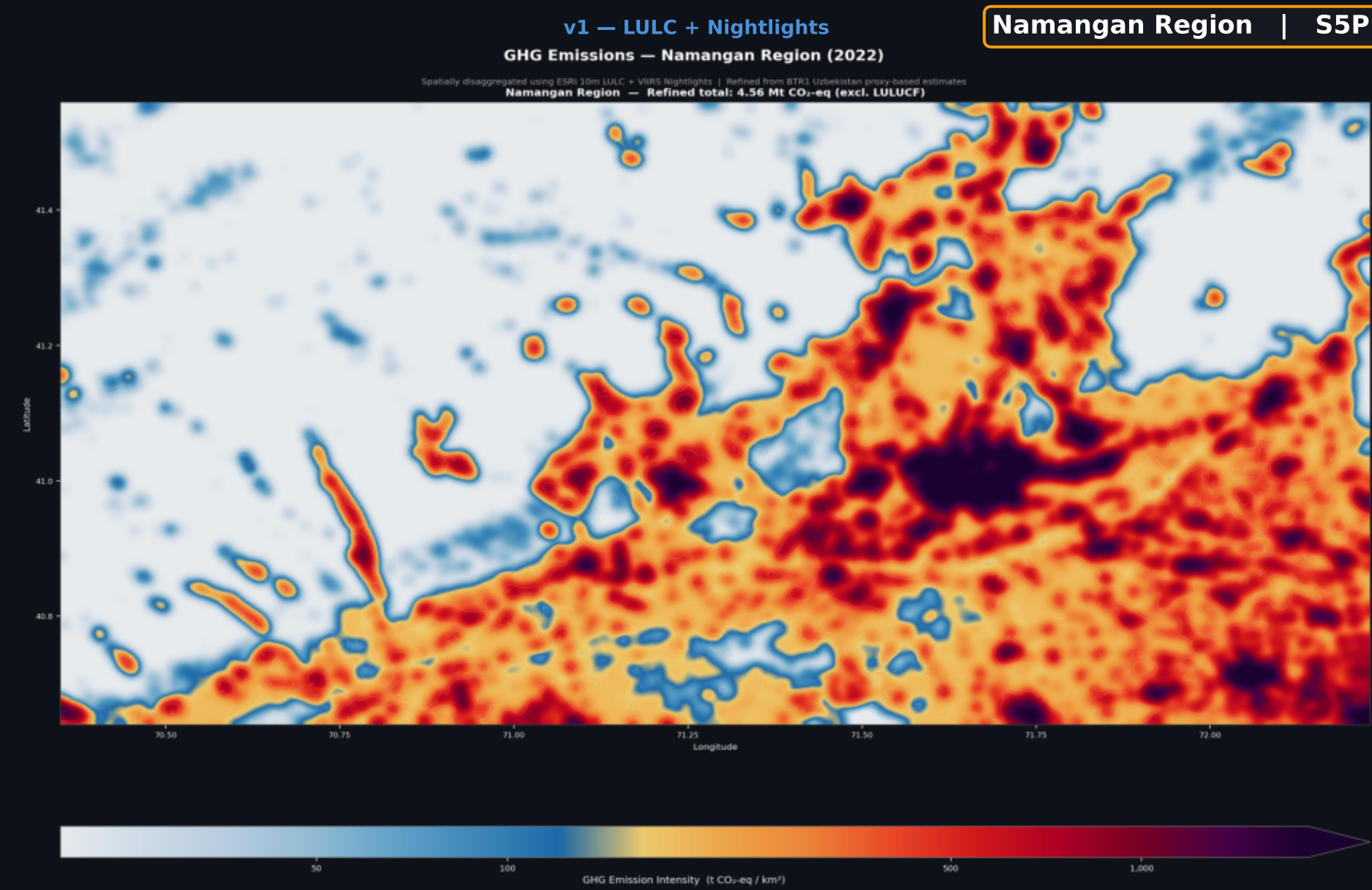
6.89 Mt CO₂-eq

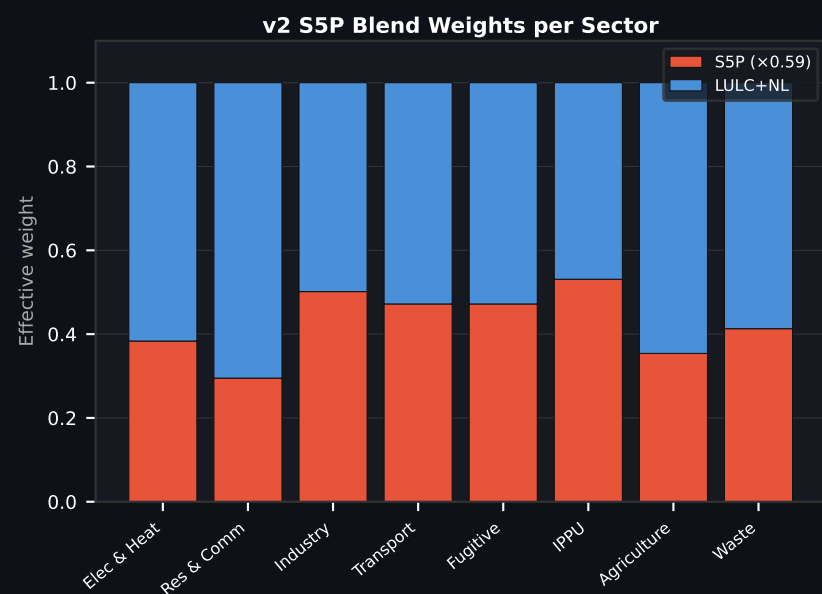
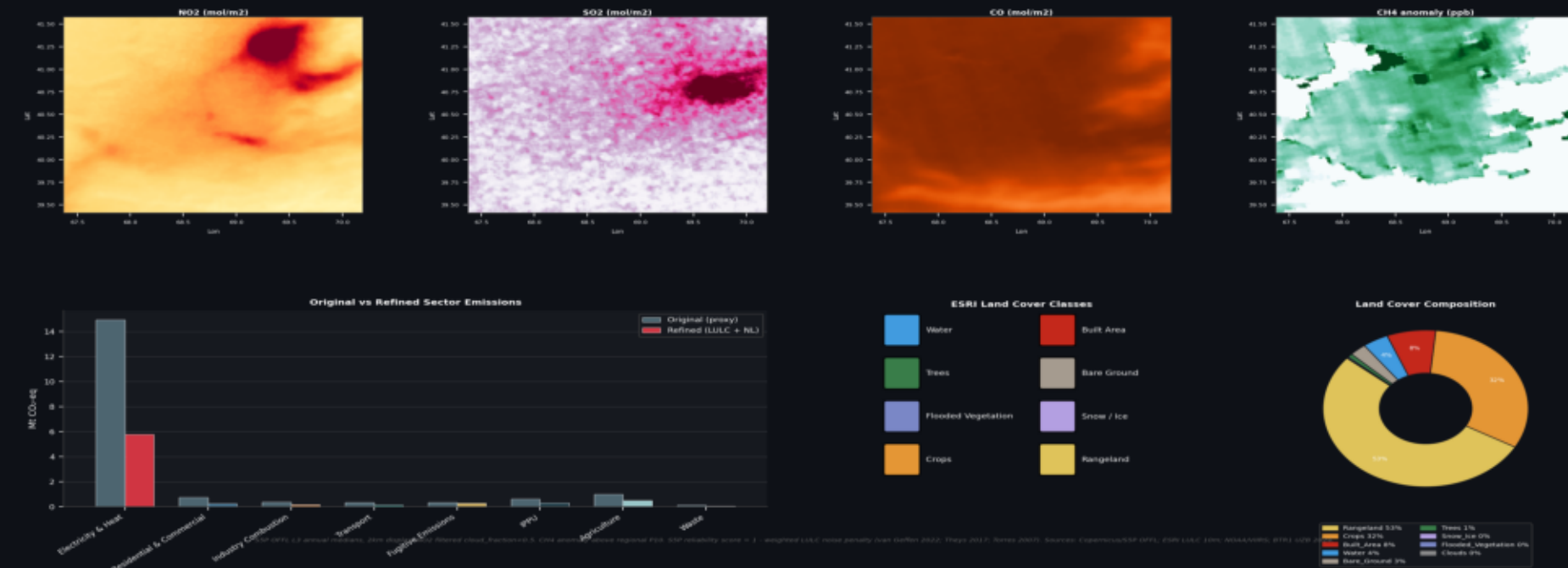
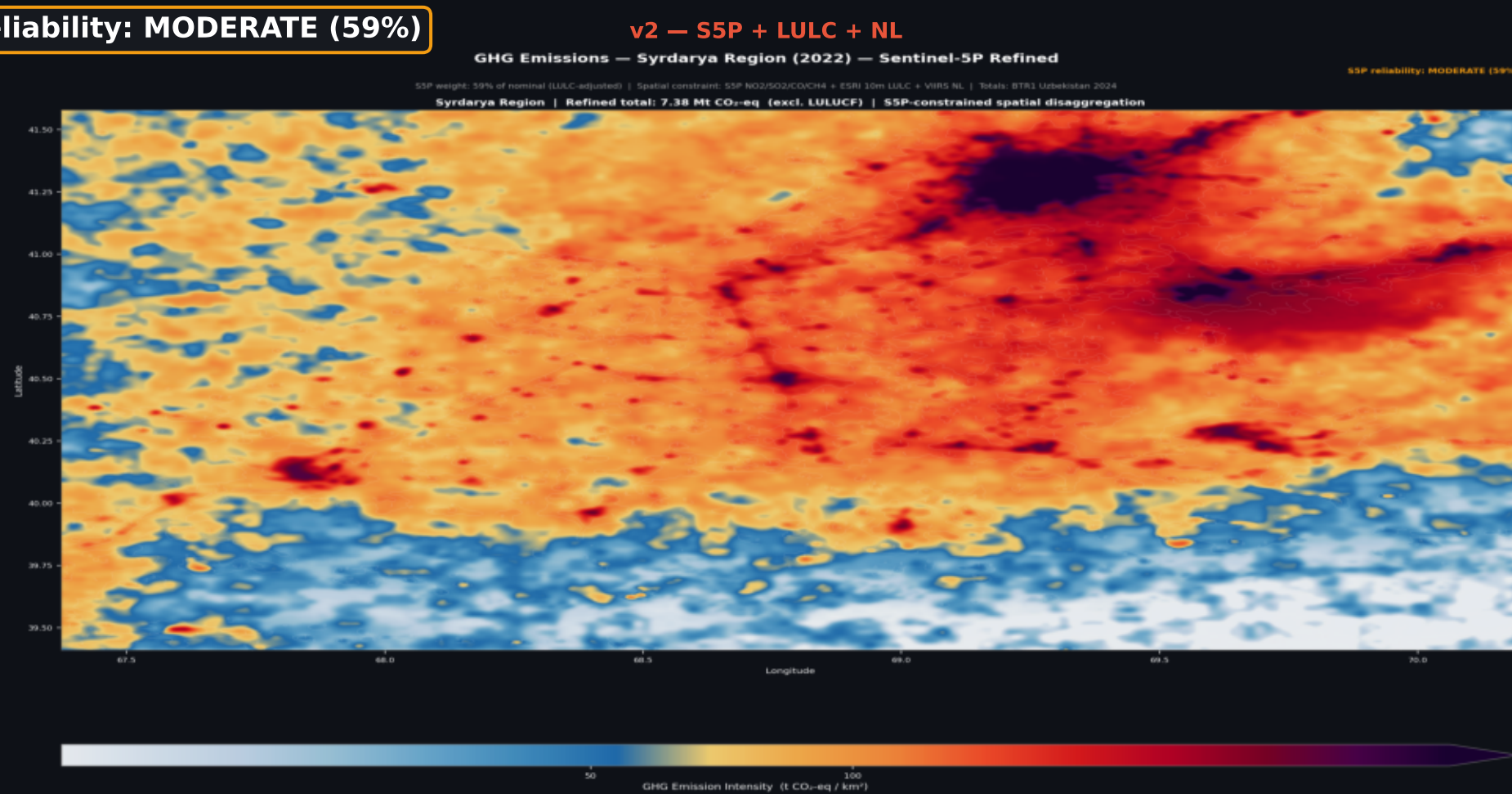
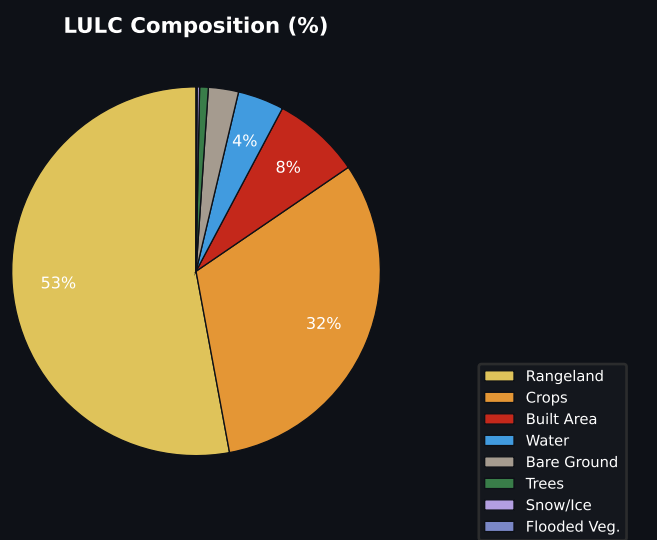
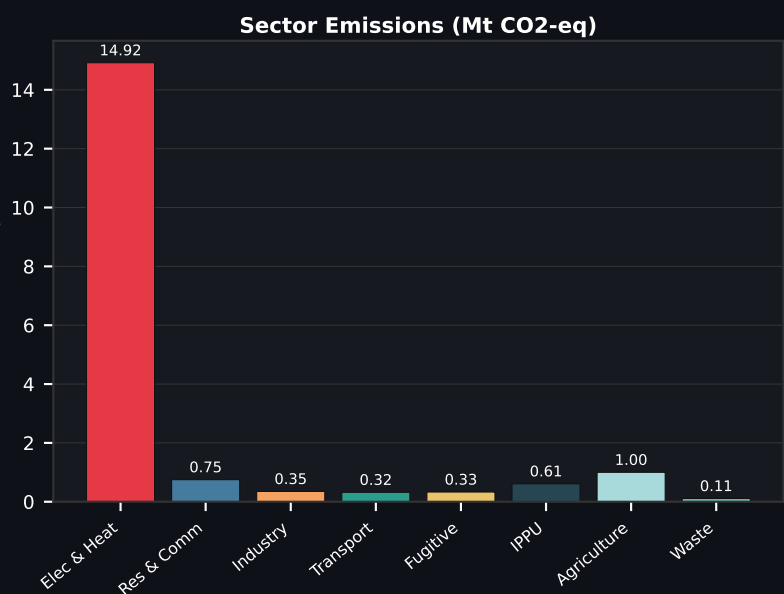
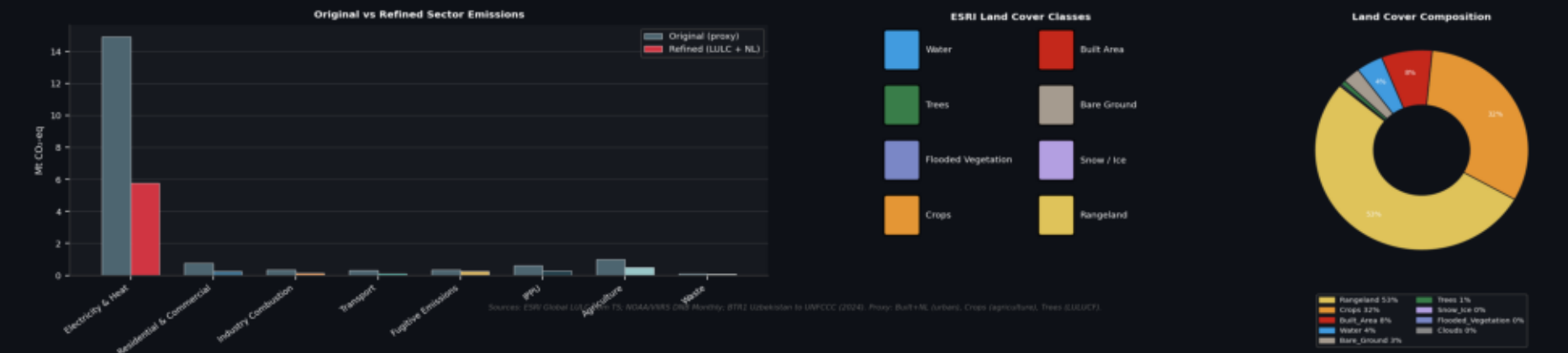
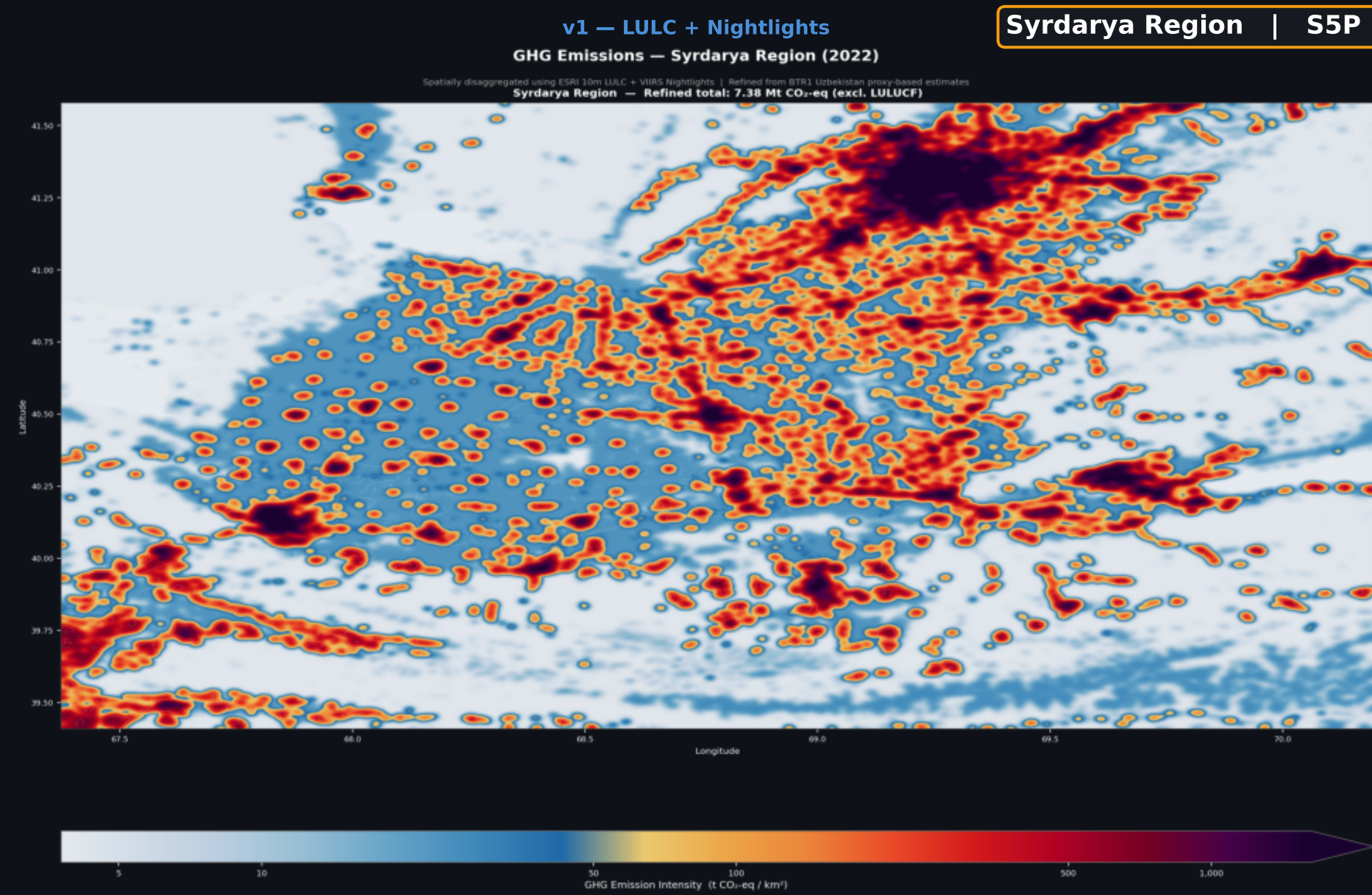
Dominant sector:

Agriculture

Dominant LULC:

Rangeland





S5P Reliability:
MODERATE (59%)

Approach:
Blended

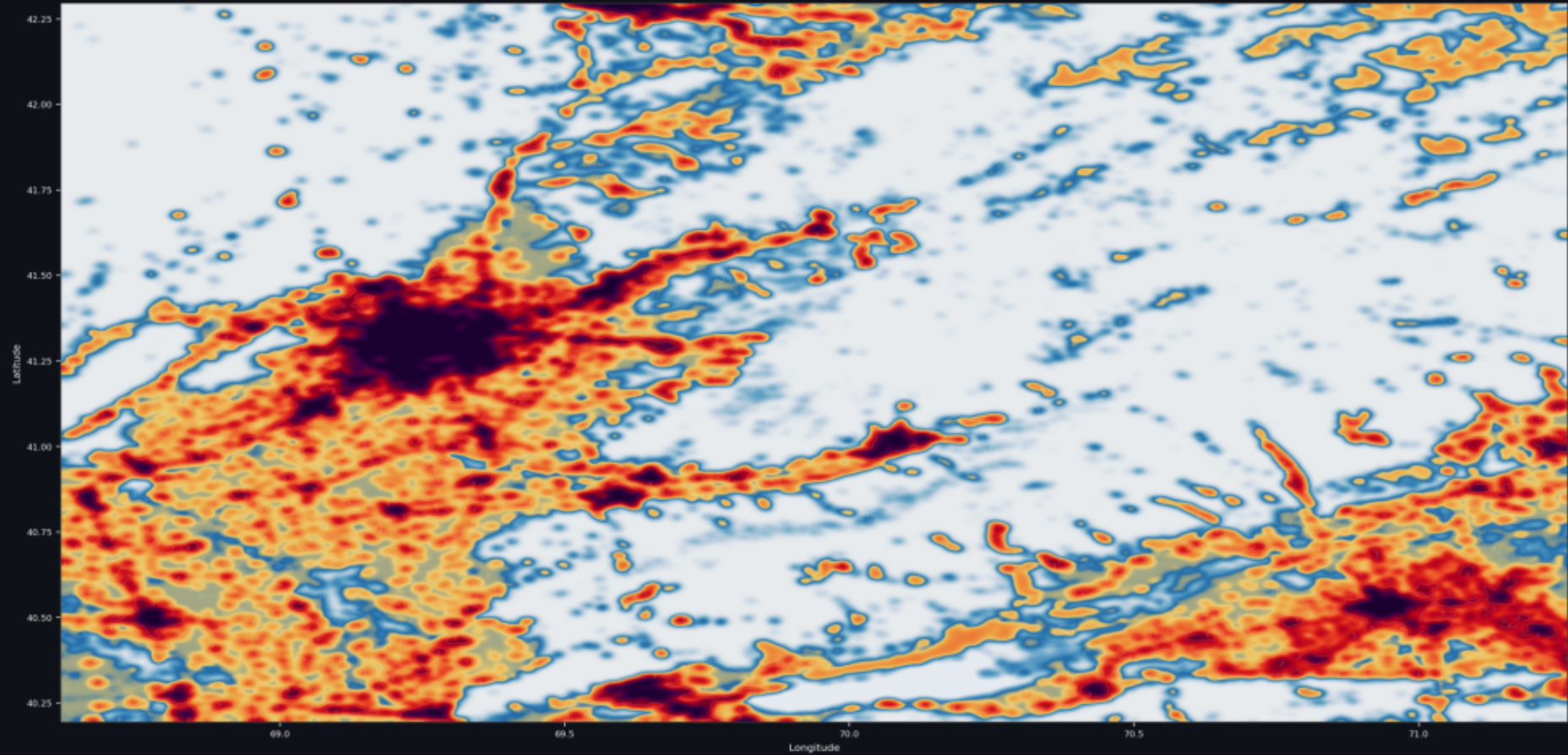
Total emissions:
18.39 Mt CO₂-eq

Dominant sector:
Electricity Heat

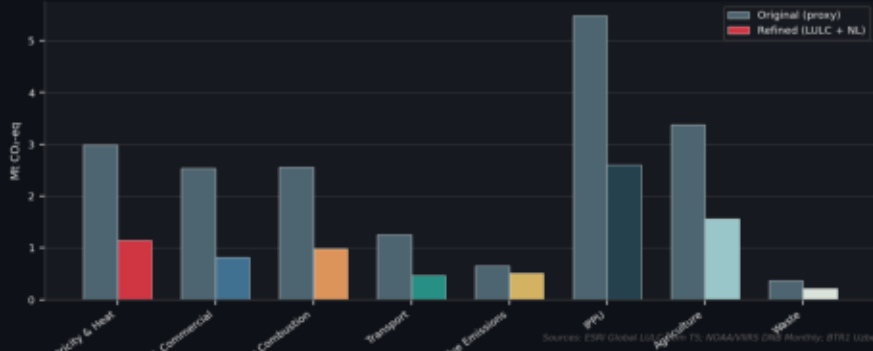
Dominant LULC:
Rangeland

GHG Emissions — Tashkent Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR1 Uzbekistan proxy-based estimates
Tashkent Region — Refined total: 8.30 Mt CO₂-eq (excl. LULUCF)



Original vs Refined Sector Emissions

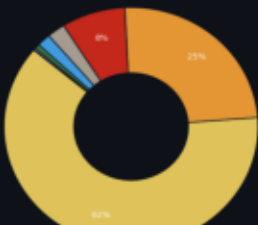


Sources: ESRI Global CO₂ Land Use (v1), NOAA/NOAA Data Inventory, BTR1 Uzbekistan to UNFCCC (2024). Proxy: Built + NL, Fugitive, Crops (agriculture), Trees (ESLUCE).

ESRI Land Cover Classes



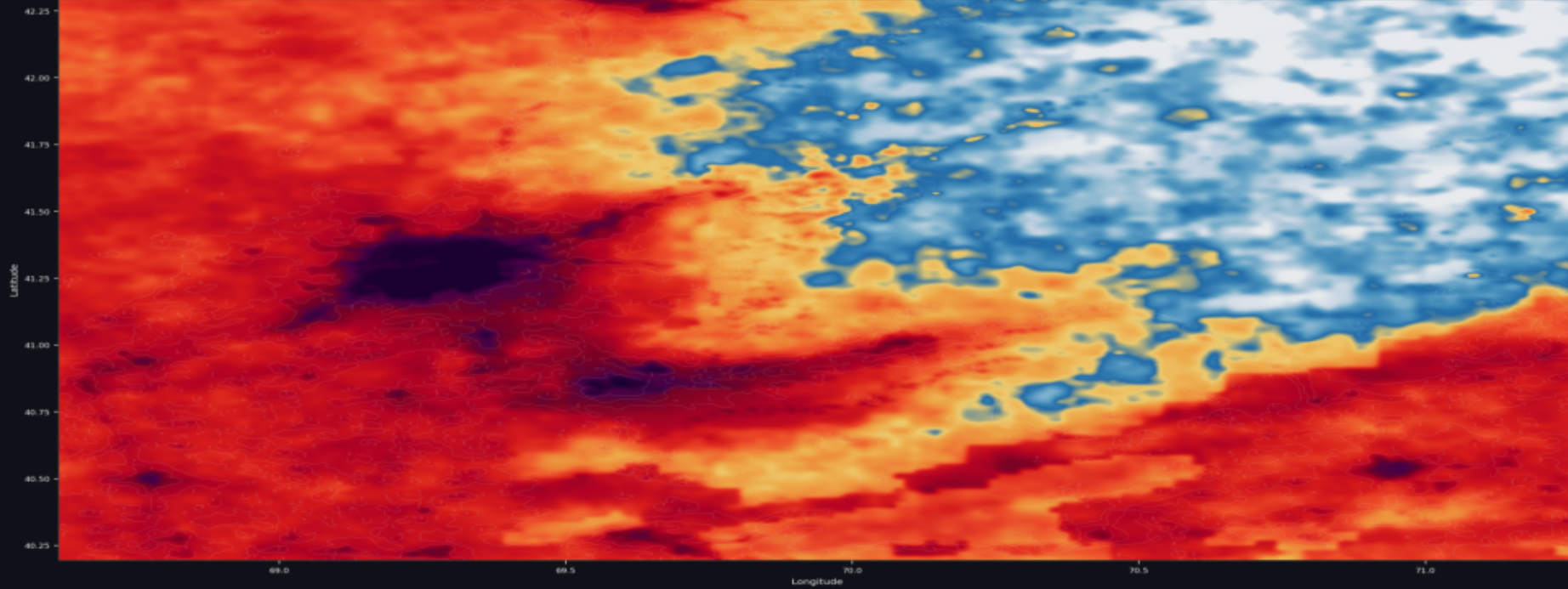
Land Cover Composition



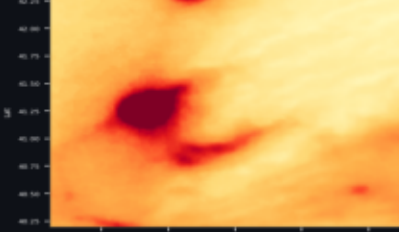
Rangeland 62%
Crops 25%
Built Area 8%
Water 2%
Bare Ground 2%
Flooded Vegetation 0%
Snow/Ice 0%
Trees 1%

GHG Emissions — Tashkent Region (2022) — Sentinel-5P Refined

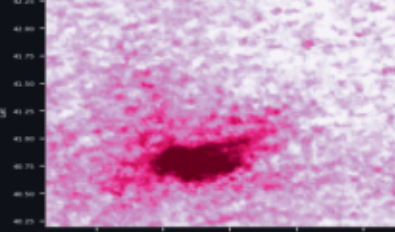
S5P weight: 54% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESRI 10m LULC + VIIRS NL | Totals: BTR1 Uzbekistan 2024
Tashkent Region | Refined total: 8.30 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation



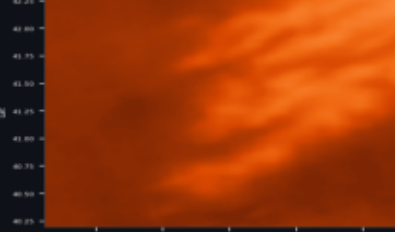
NO2 (mol/m2)



SO2 (mol/m2)



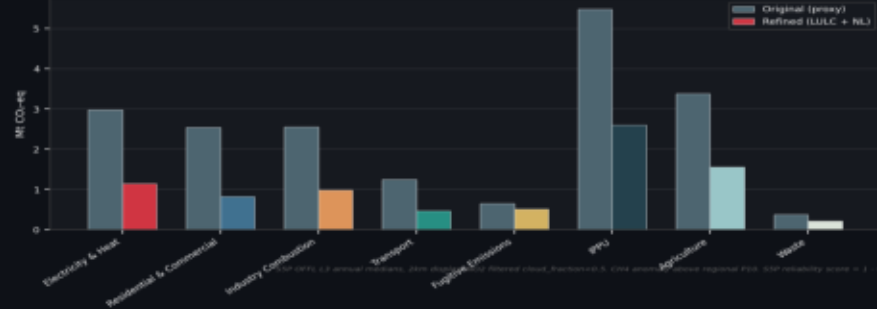
CO (mol/m2)



CH4 anomaly (ppb)



Original vs Refined Sector Emissions

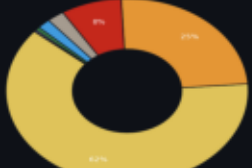


Sources: ESRI Global CO₂ Land Use (v1), NOAA/NOAA Data Inventory, BTR1 Uzbekistan to UNFCCC (2024). Proxy: Built + NL, Fugitive, Crops (agriculture), Trees (ESLUCE).

ESRI Land Cover Classes

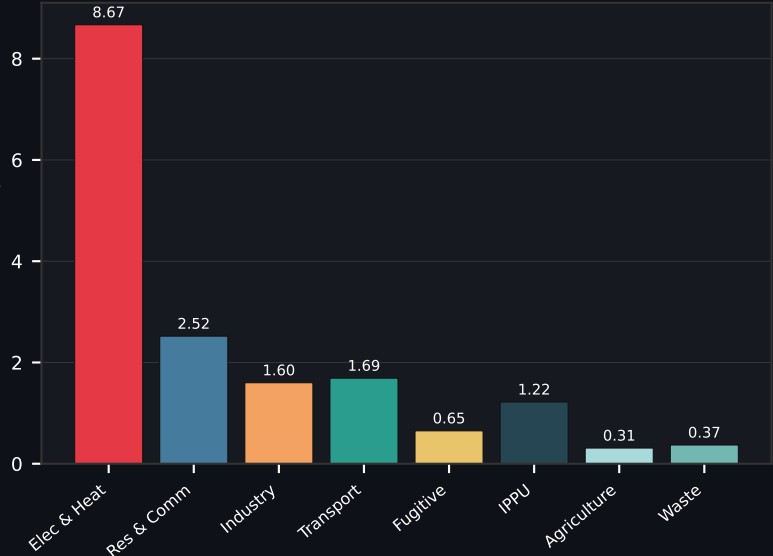


Land Cover Composition

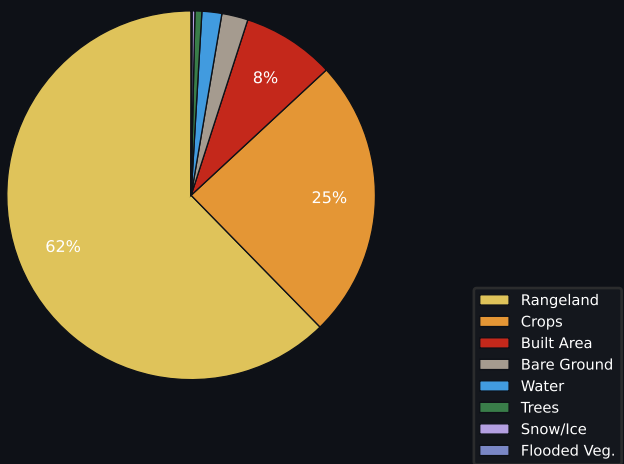


Rangeland 62%
Crops 25%
Built Area 8%
Water 2%
Bare Ground 2%
Flooded Vegetation 0%
Snow/Ice 0%
Trees 1%

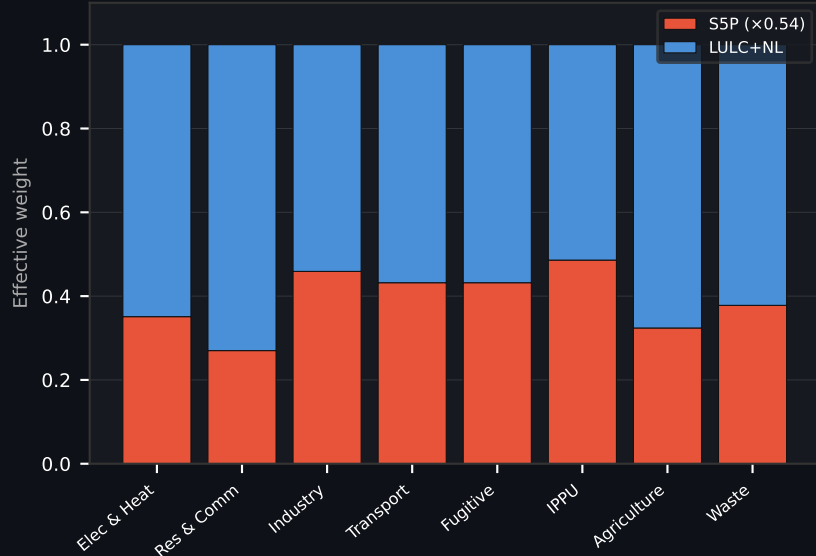
Sector Emissions (Mt CO₂-eq)



LULC Composition (%)



v2 S5P Blend Weights per Sector



S5P Reliability:

MODERATE (54%)

Approach:

Blended

Total emissions:

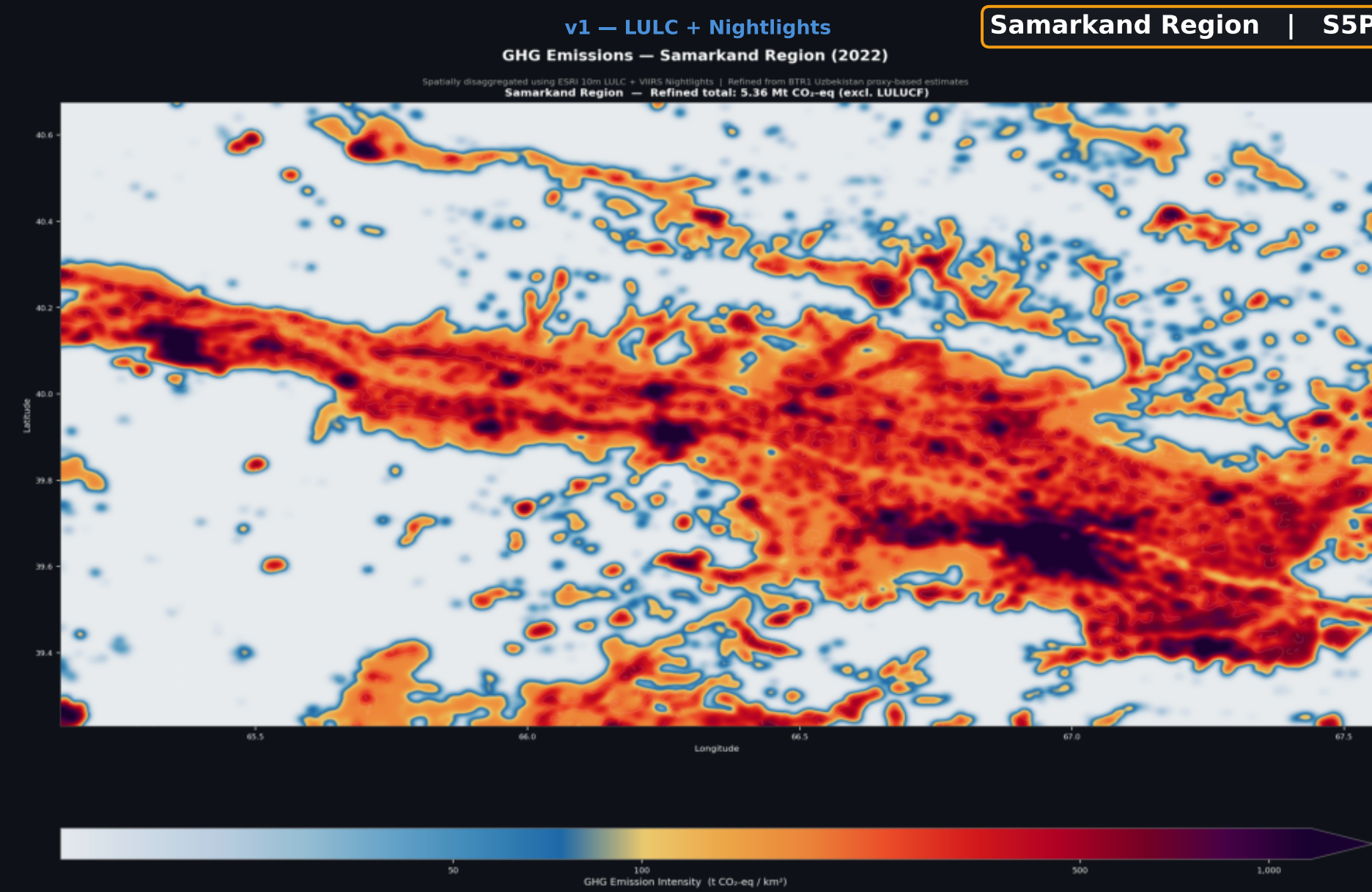
17.03 Mt CO₂-eq

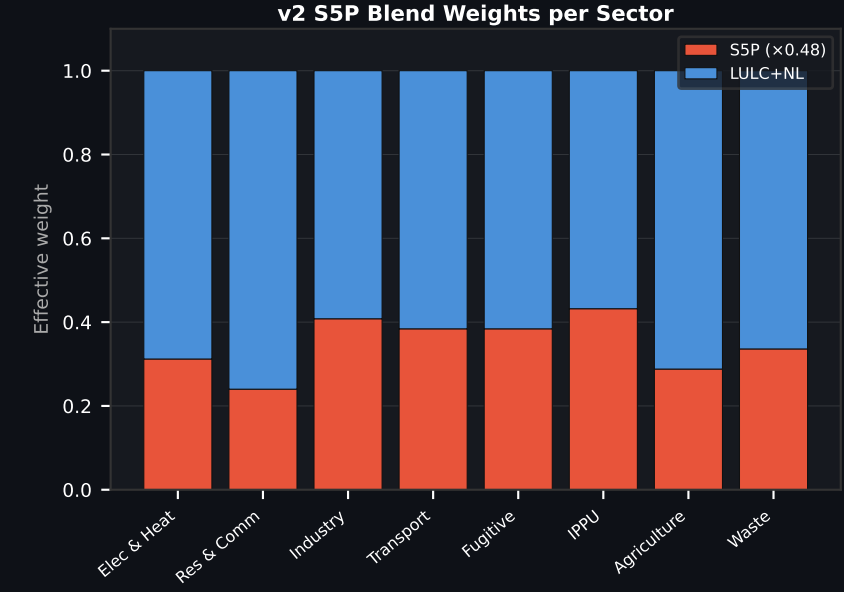
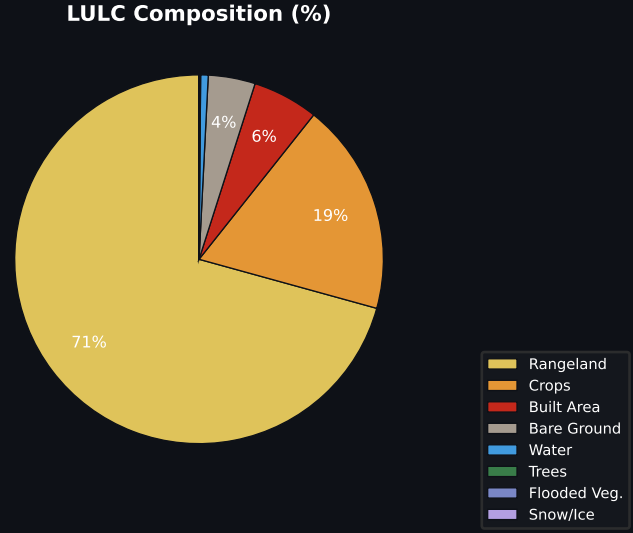
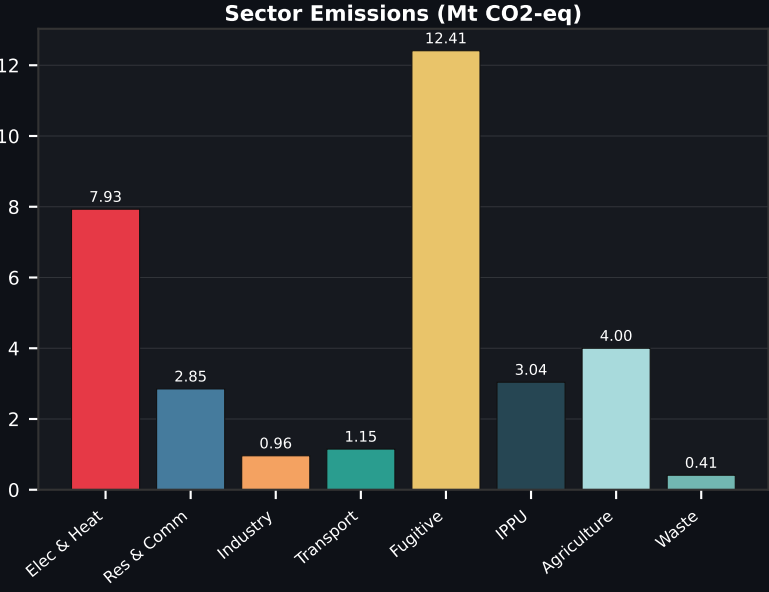
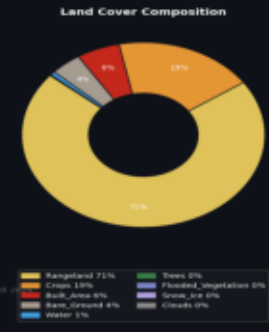
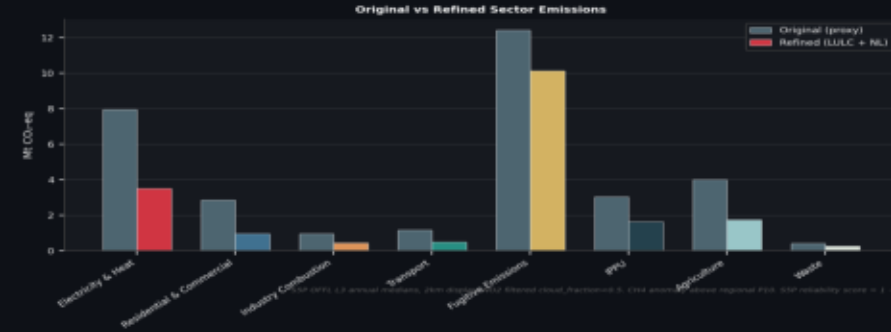
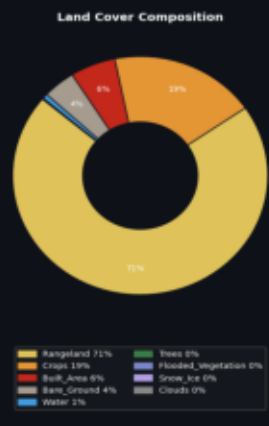
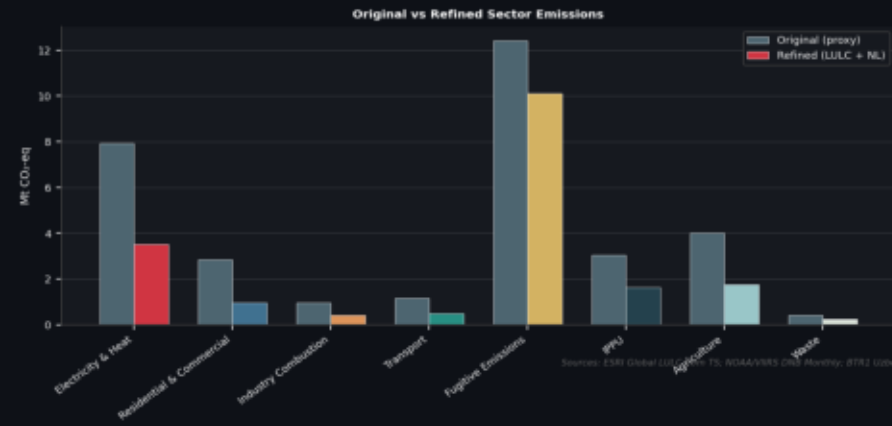
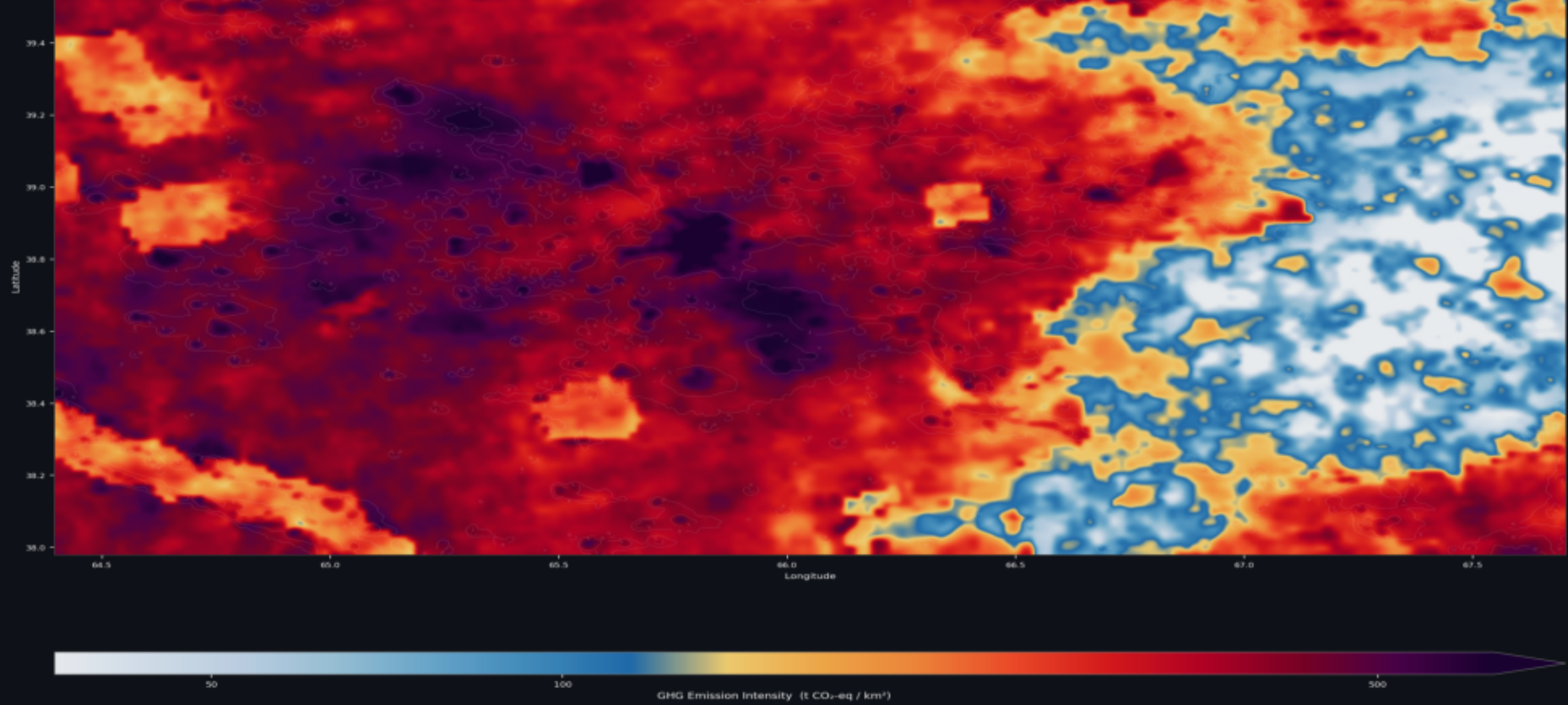
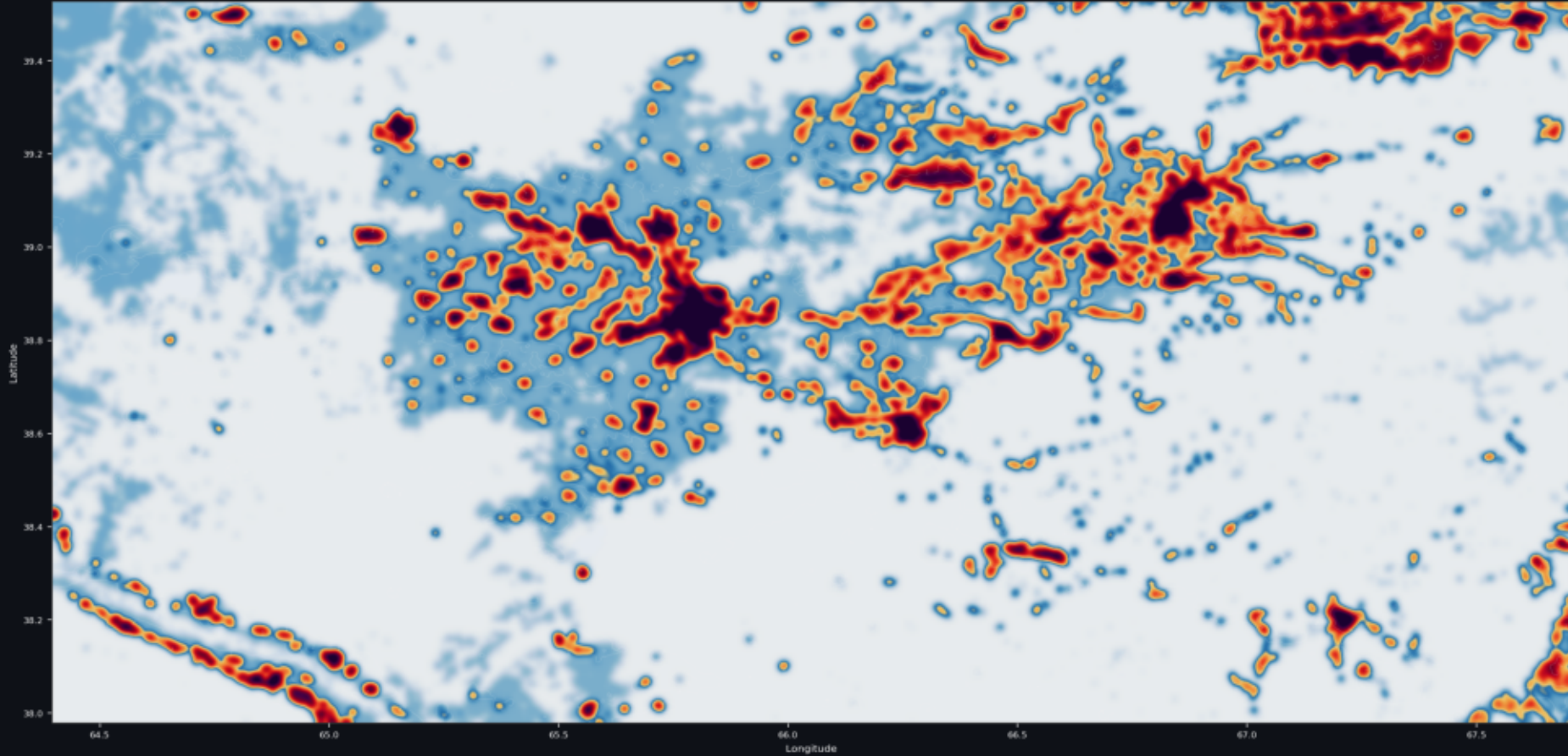
Dominant sector:

Electricity Heat

Dominant LULC:

Rangeland



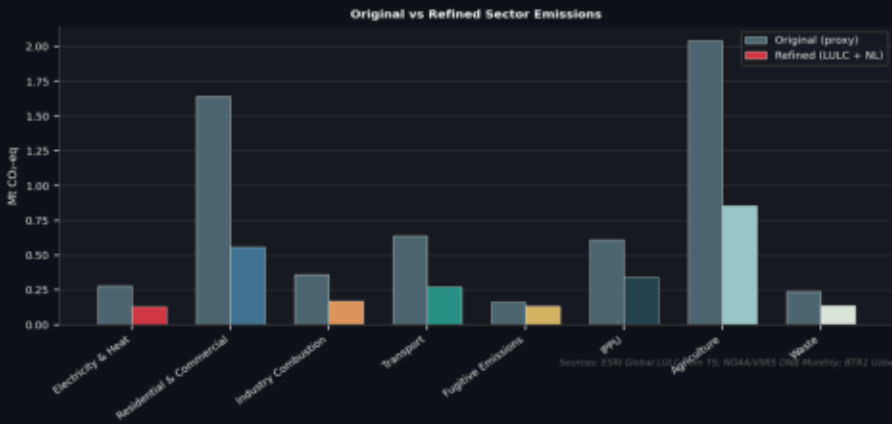
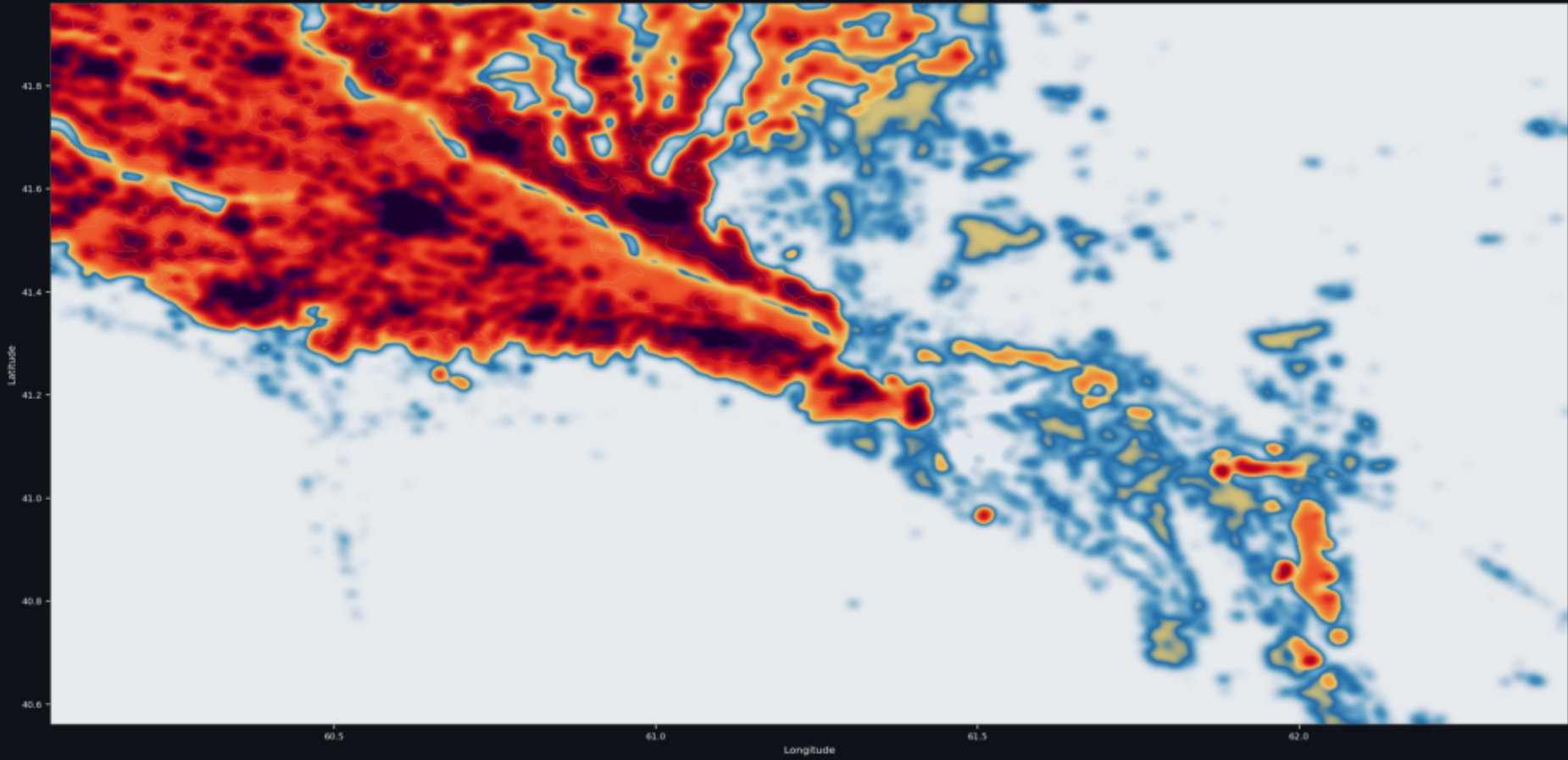


S5P Reliability:
MODERATE (48%)
Approach:
Blended
Total emissions:
32.75 Mt CO₂-eq
Dominant sector:
Fugitive Emissions
Dominant LULC:
Rangeland

v1 — LULC + Nightlights

GHG Emissions — Khorezm Region (2022)

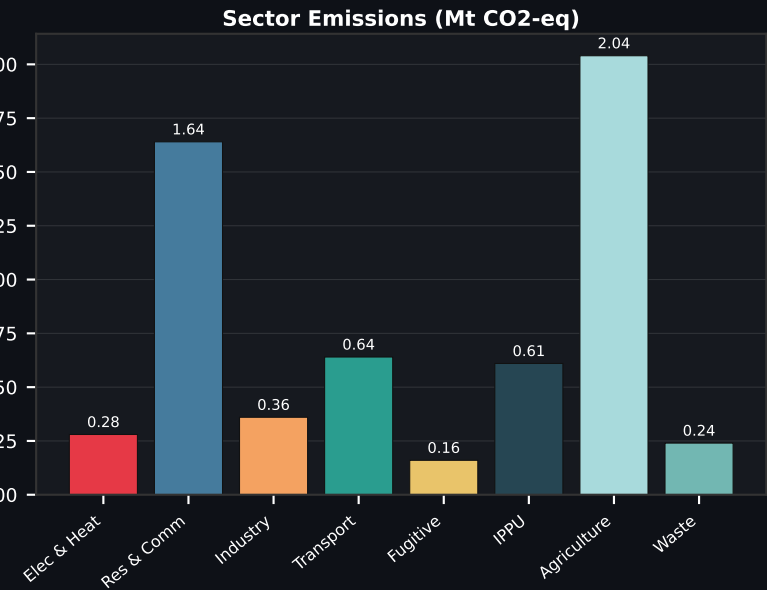
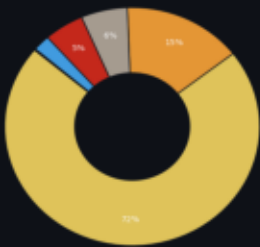
Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR3 Uzbekistan proxy-based estimates
Khorezm Region — Refined total: 2.59 Mt CO₂-eq (excl. LULUCF)



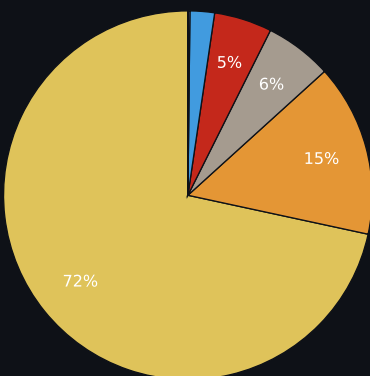
ESRI Land Cover Classes



Land Cover Composition



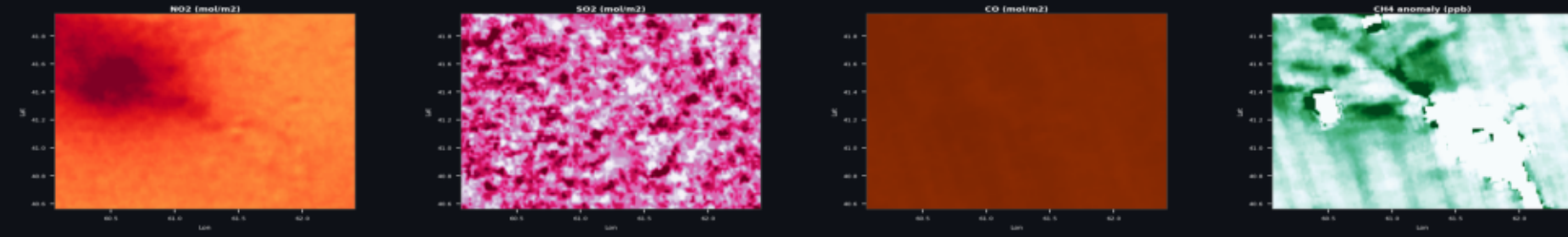
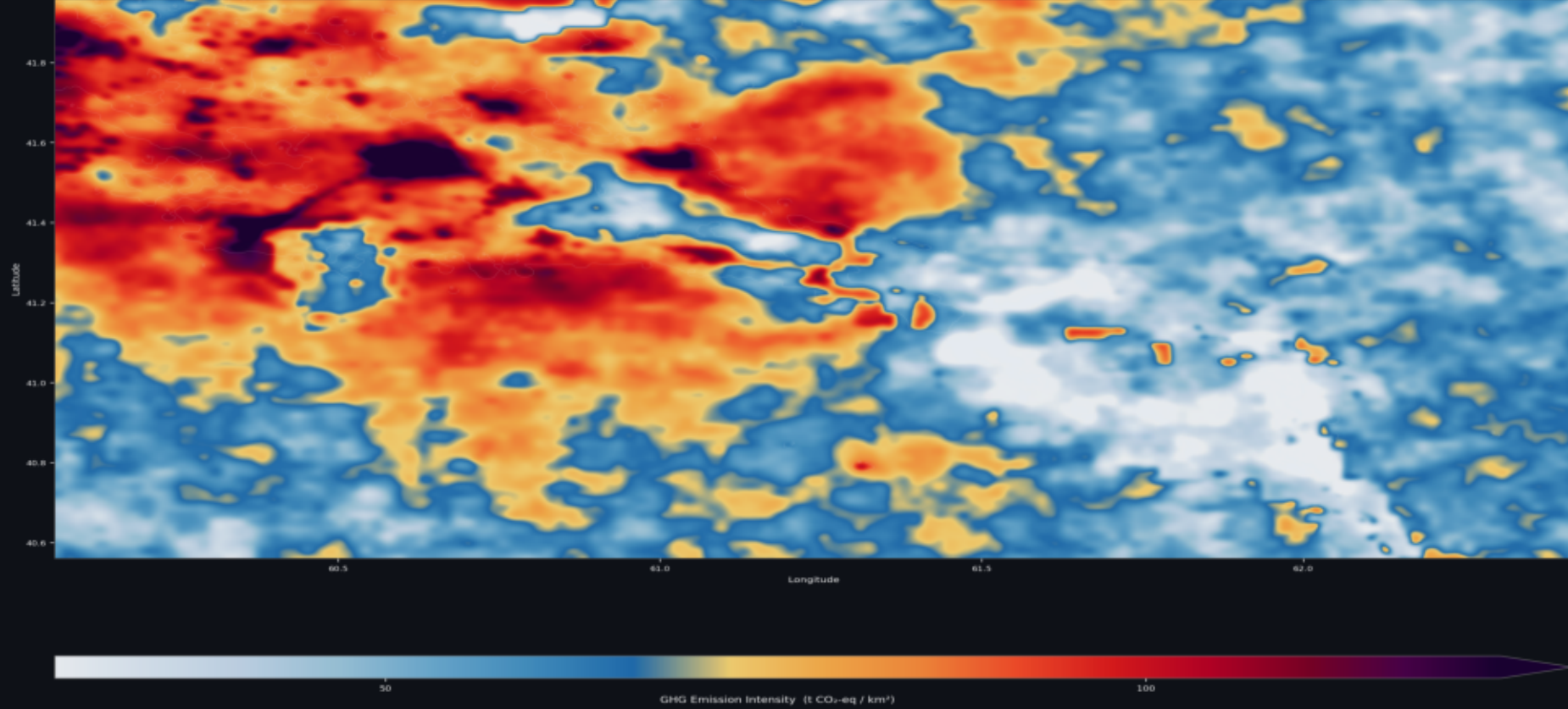
LULC Composition (%)



v2 — S5P + LULC + NL

GHG Emissions — Khorezm Region (2022) — Sentinel-5P Refined

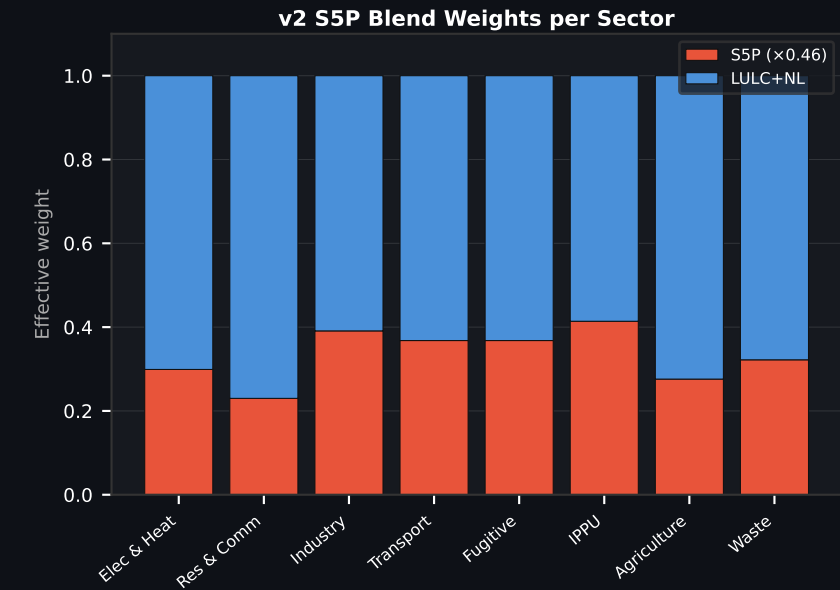
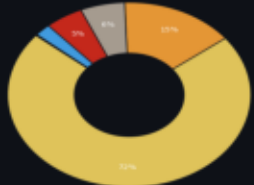
S5P weight: 46% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESRI 10m LULC + VIIRS NL | Totals: BTR3 Uzbekistan 2024
Khorezm Region | Refined total: 2.59 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation



ESRI Land Cover Classes



Land Cover Composition



S5P Reliability:

MODERATE (46%)

Approach:

Blended

Total emissions:

5.97 Mt CO₂-eq

Dominant sector:

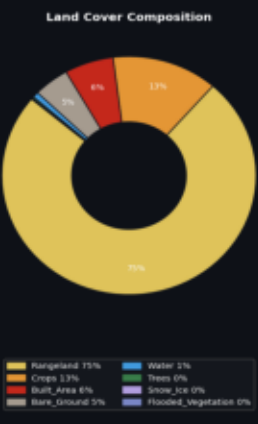
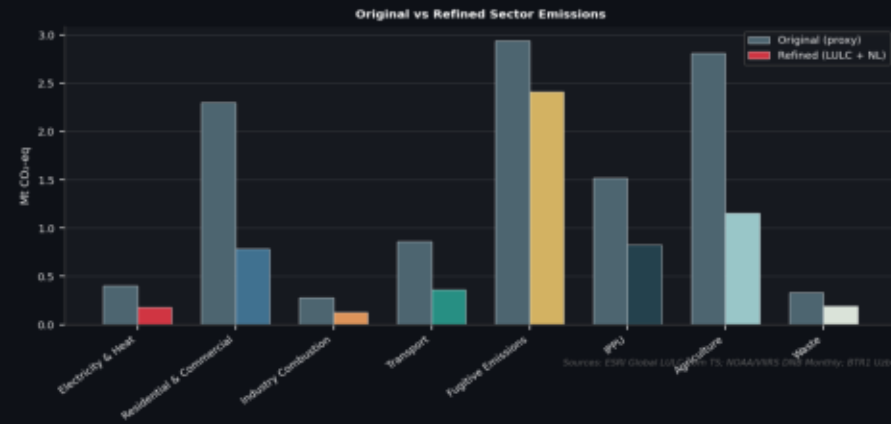
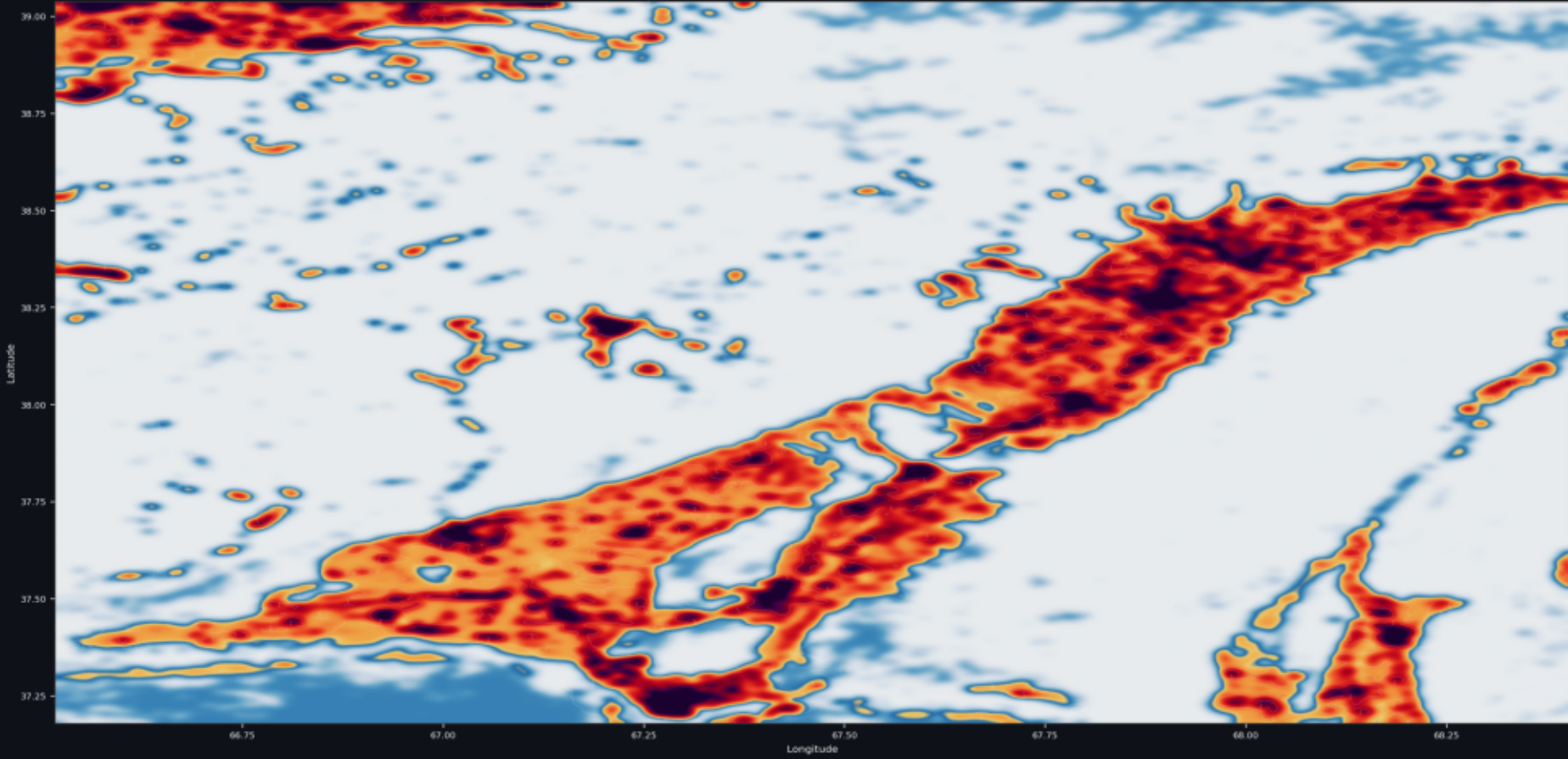
Agriculture

Dominant LULC:

Rangeland

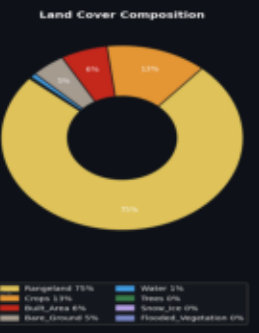
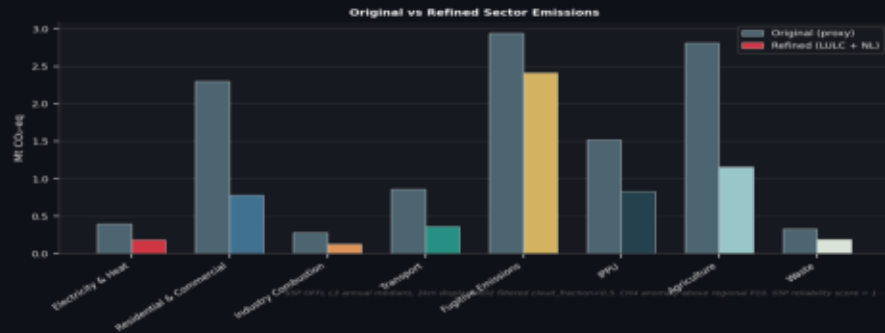
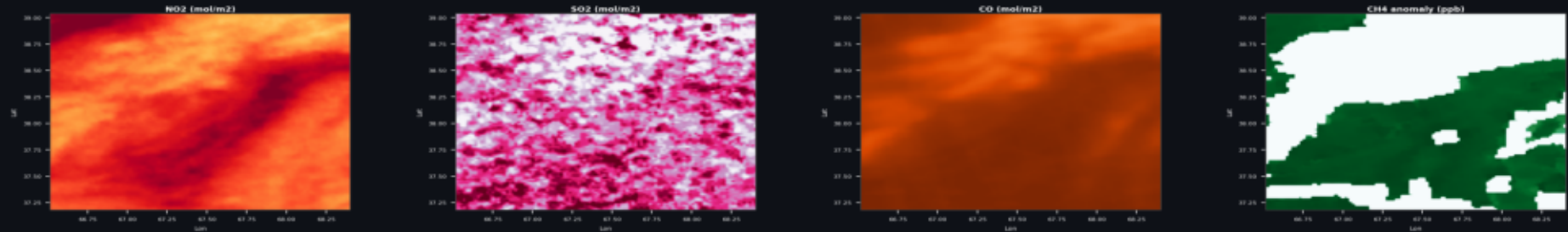
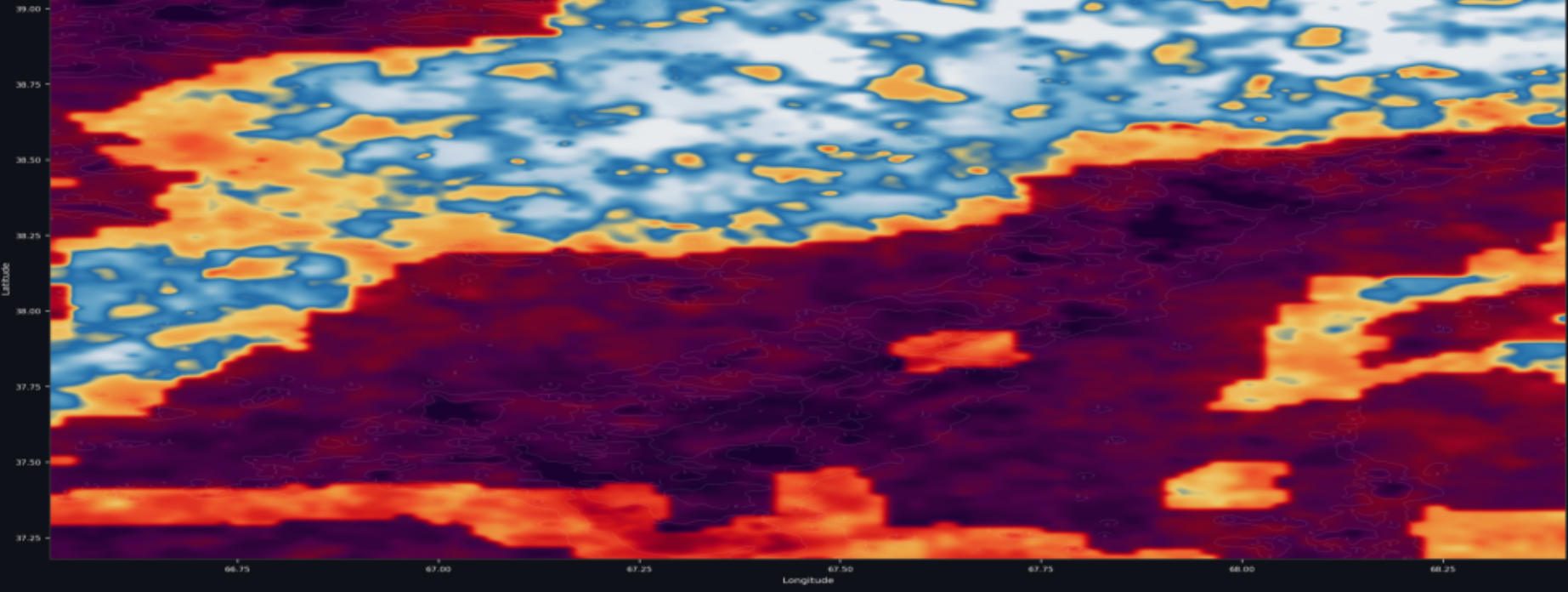
GHG Emissions — Surkhandarya Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR1 Uzbekistan proxy-based estimates
Surkhandarya Region — Refined total: 6.03 Mt CO₂-eq (excl. LULUCF)

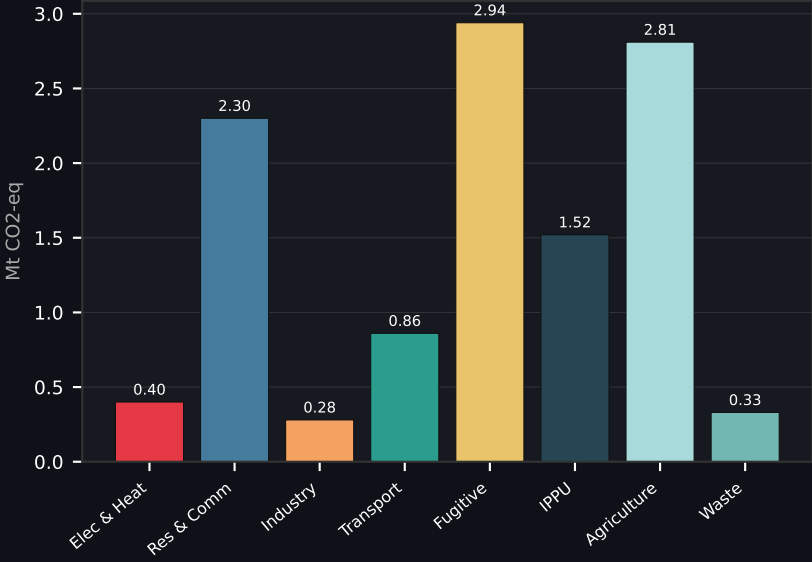


GHG Emissions — Surkhandarya Region (2022) — Sentinel-5P Refined

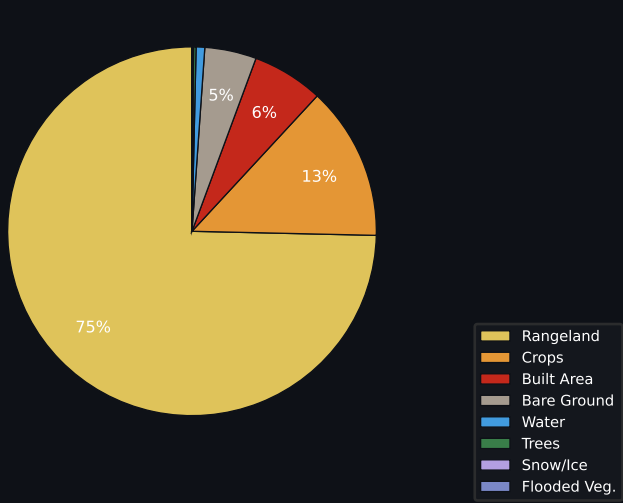
S5P weight: 45% of nominal (LULC-adjusted) | Spatial constraint: S5P NO₂/SO₂/CO/CH₄ + ESRI 10m LULC + VIIRS NL | Totals: BTR1 Uzbekistan 2024
Surkhandarya Region | Refined total: 6.03 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation



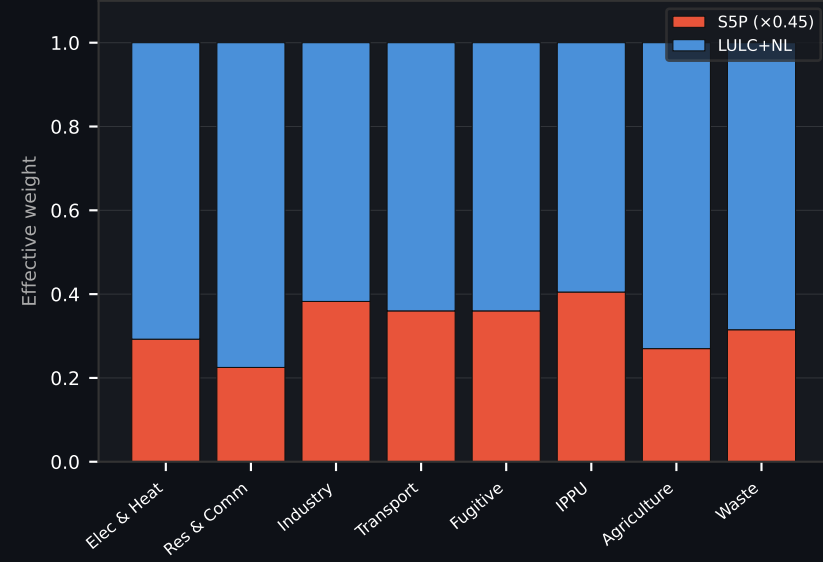
Sector Emissions (Mt CO₂-eq)



LULC Composition (%)



v2 S5P Blend Weights per Sector



S5P Reliability:

MODERATE (45%)

Approach:

Blended

Total emissions:

11.44 Mt CO₂-eq

Dominant sector:

Fugitive Emissions

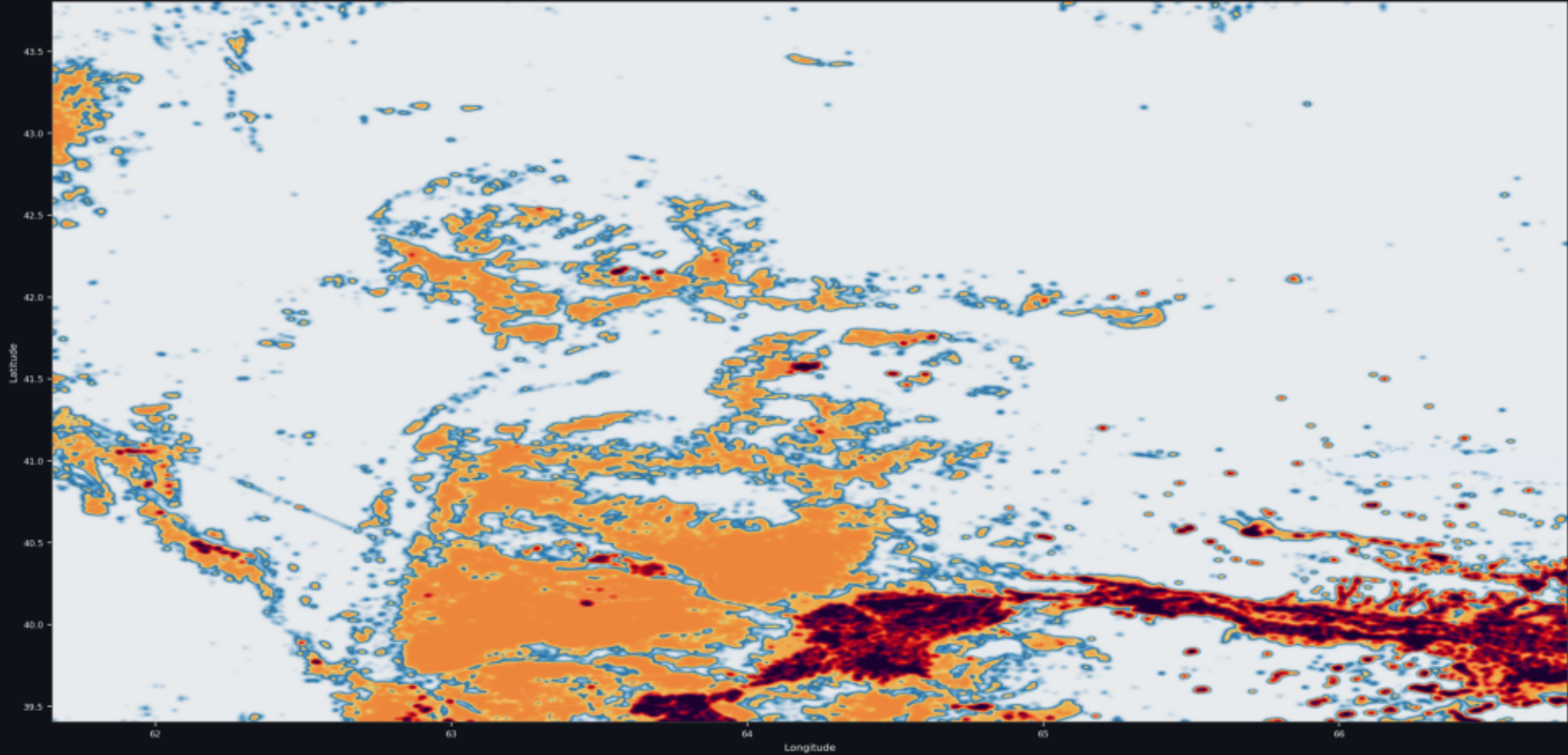
Dominant LULC:

Rangeland

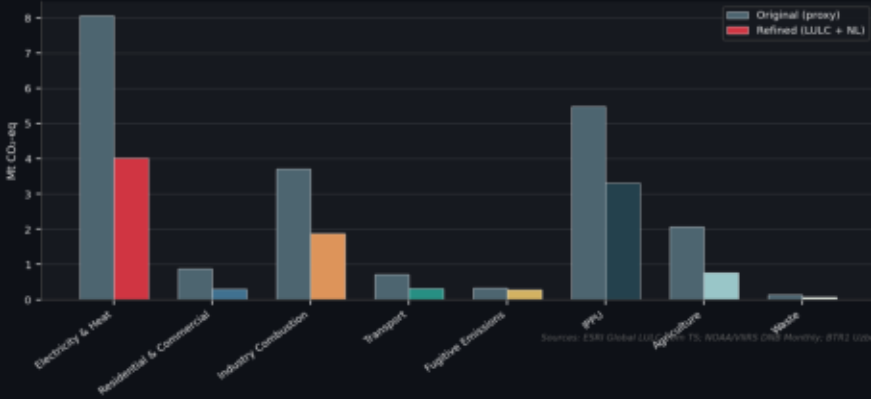
v1 — LULC + Nightlights

GHG Emissions — Navoi Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR3 Uzbekistan proxy-based estimates
Navoi Region — Refined total: 10.94 Mt CO₂-eq (excl. LULUCF)



Original vs Refined Sector Emissions

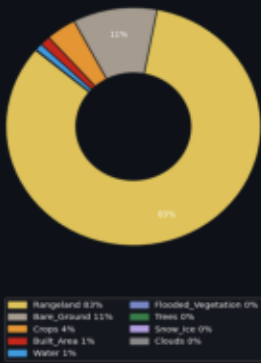


Source: ESRI Global LULC v2020 10m, NOAA VIIRS DMSP Nightlights, BTR3 Uzbekistan to UNFCCC (2024), Proxy: Built+NL, Landuse, Cropland, Vegetation, Trees, LULUCF

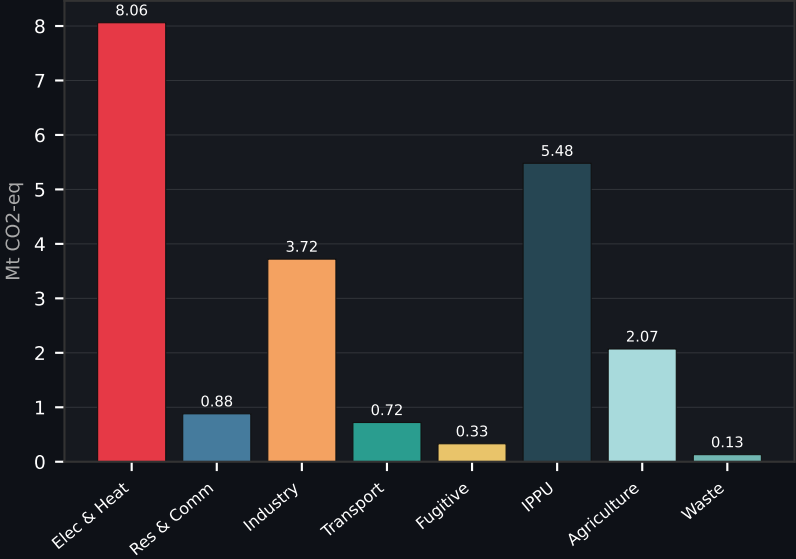
ESRI Land Cover Classes



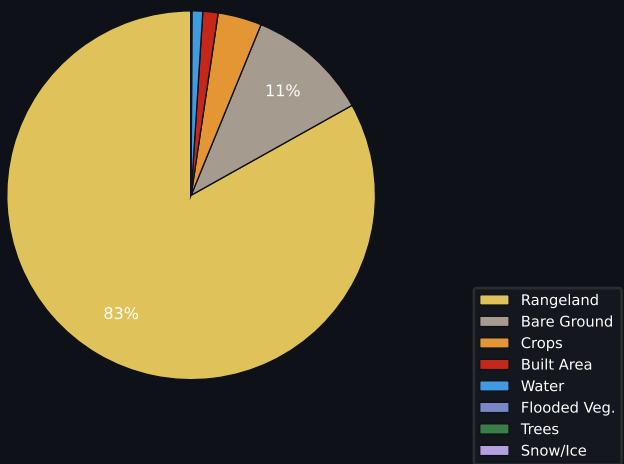
Land Cover Composition



Sector Emissions (Mt CO₂-eq)



LULC Composition (%)

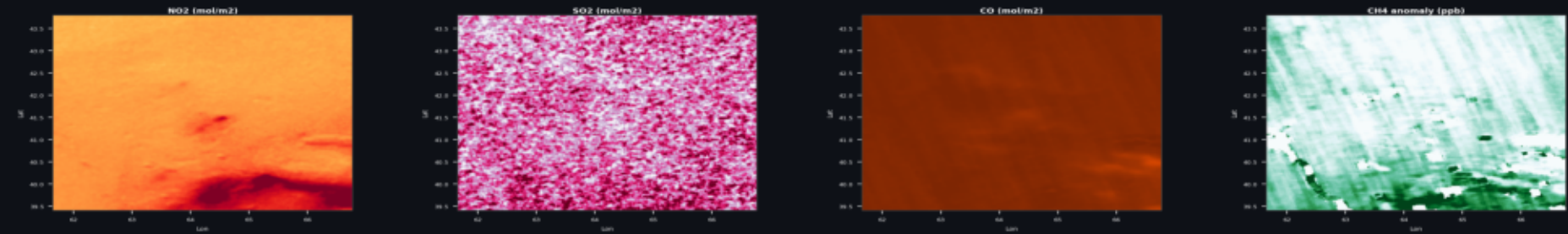
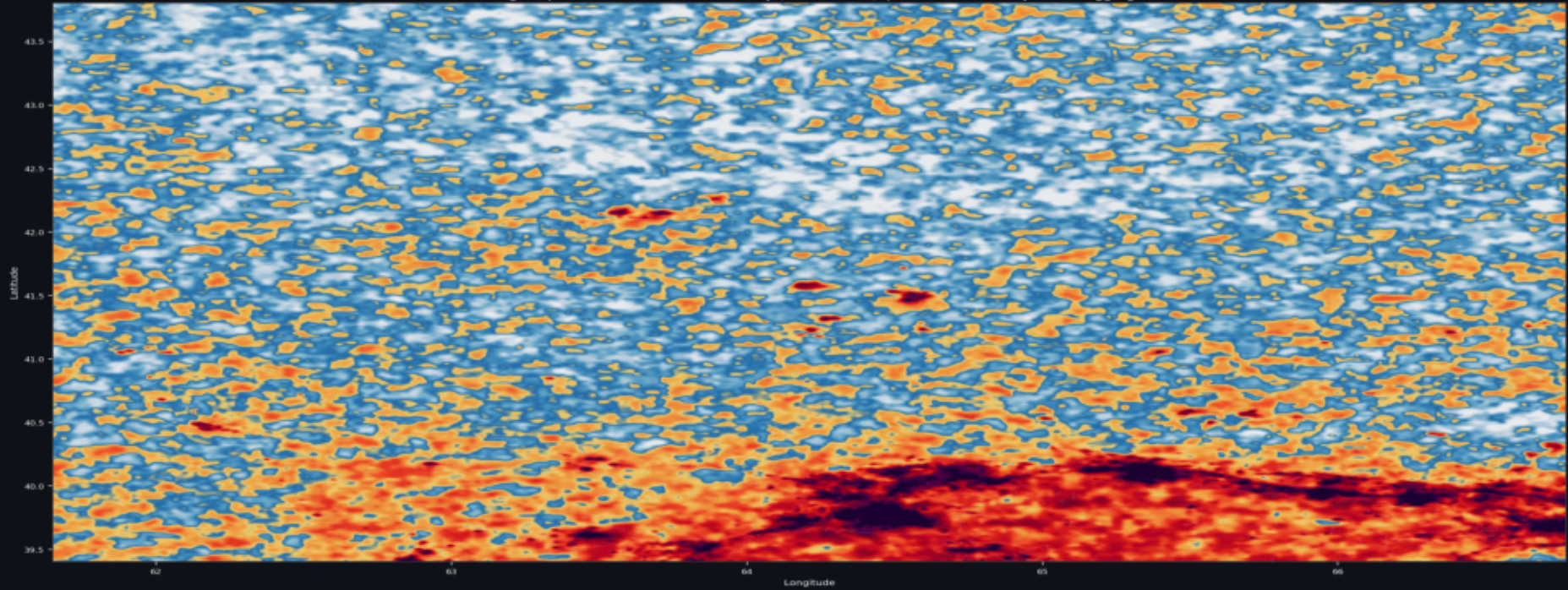


v2 — S5P + LULC + NL

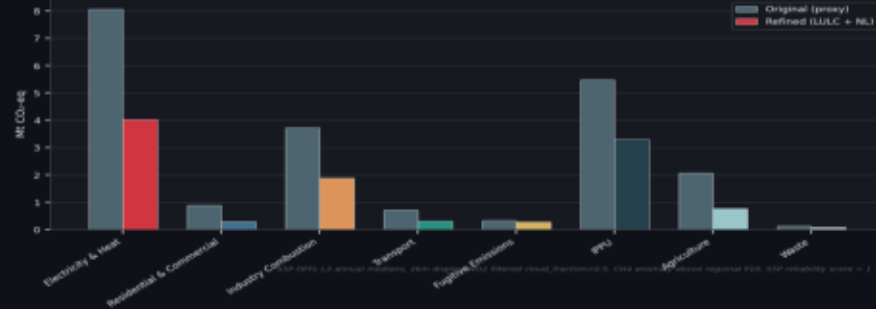
GHG Emissions — Navoi Region (2022) — Sentinel-5P Refined

S5P weight: 35% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESRI 10m LULC + VIIRS NL | Total: BTR3 Uzbekistan 2024
Navoi Region | Refined total: 10.94 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation

S5P reliability: LOW (35%)



Original vs Refined Sector Emissions

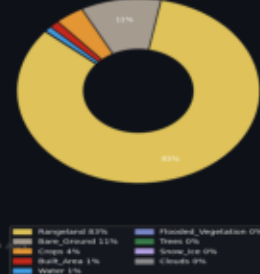


Source: ESRI Global LULC v2020 10m, NOAA VIIRS DMSP Nightlights, BTR3 Uzbekistan to UNFCCC (2024), Proxy: Built+NL, Landuse, Cropland, Vegetation, Trees, LULUCF

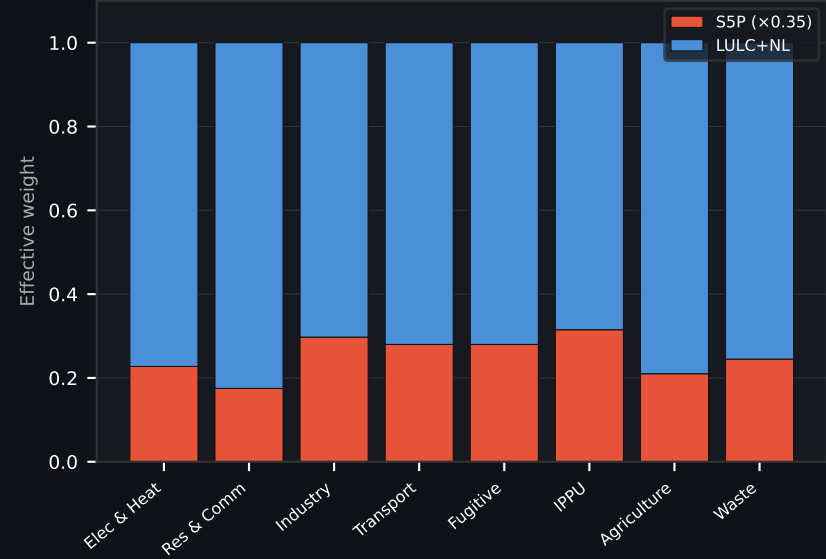
ESRI Land Cover Classes



Land Cover Composition



v2 S5P Blend Weights per Sector



S5P Reliability:

LOW (35%)

Approach:

LULC/NL dominant

Total emissions:

21.39 Mt CO₂-eq

Dominant sector:

Electricity Heat

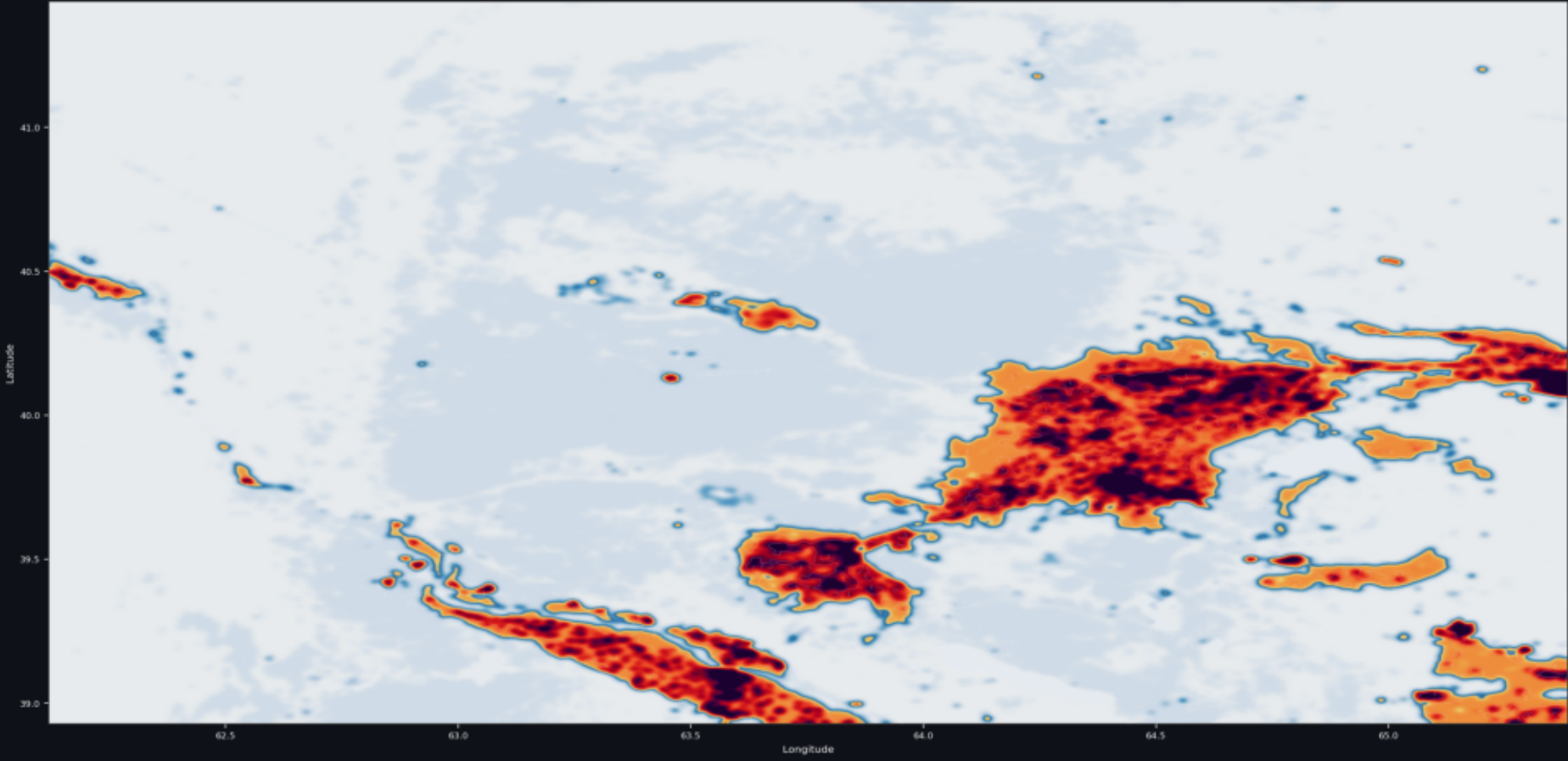
Dominant LULC:

Rangeland

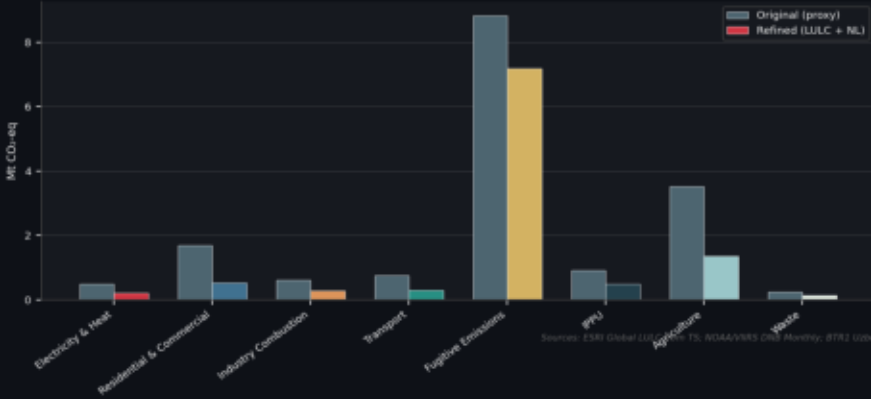
v1 — LULC + Nightlights

GHG Emissions — Bukhara Region (2022)

Spatially disaggregated using ESRI 10m LULC + VIIRS Nightlights | Refined from BTR3 Uzbekistan proxy-based estimates
Bukhara Region — Refined total: 10.47 Mt CO₂-eq (excl. LULUCF)



Original vs Refined Sector Emissions

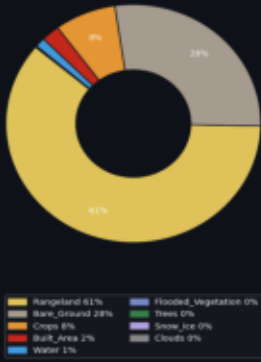


Source: ESRI Global LULC v2020 10m, NOAA VIIRS DMSP Monthly, BTR3 Uzbekistan to UNEPCE (2024), Proxy: Built+NL, Industrial, Crops (aggregations), Trees (LULUCF)

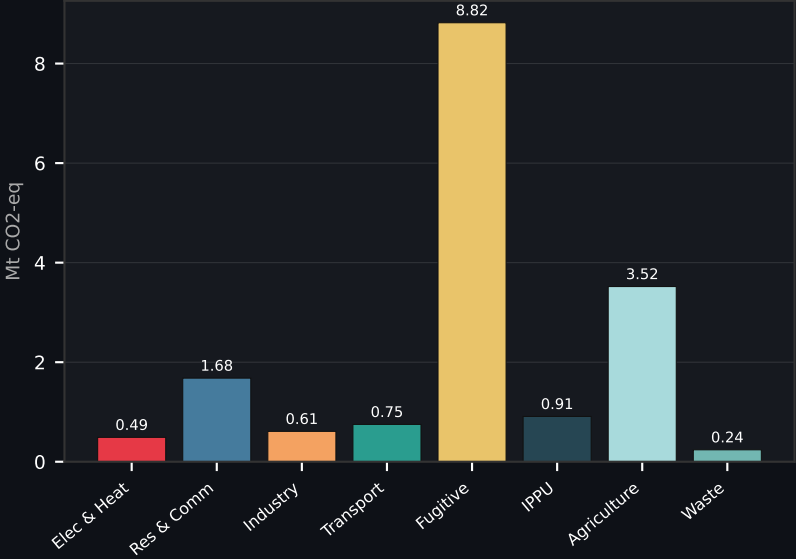
ESRI Land Cover Classes



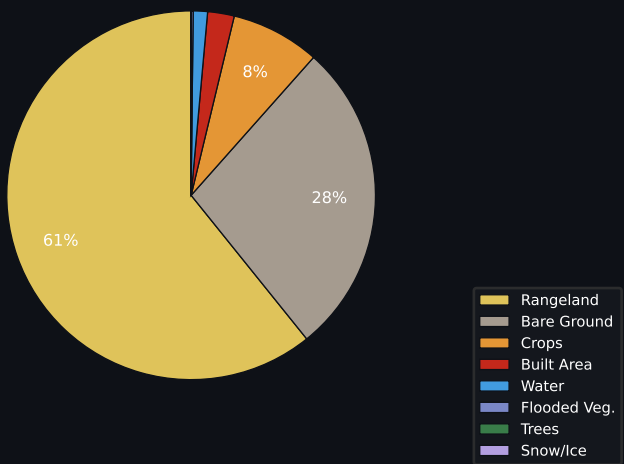
Land Cover Composition



Sector Emissions (Mt CO₂-eq)



LULC Composition (%)

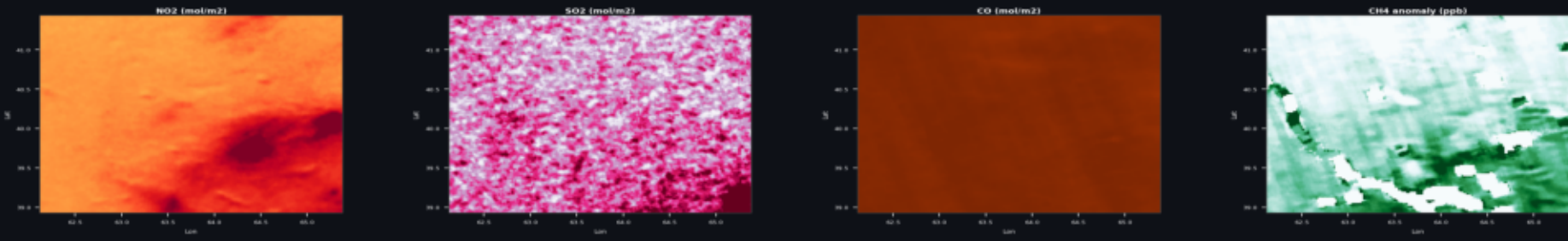
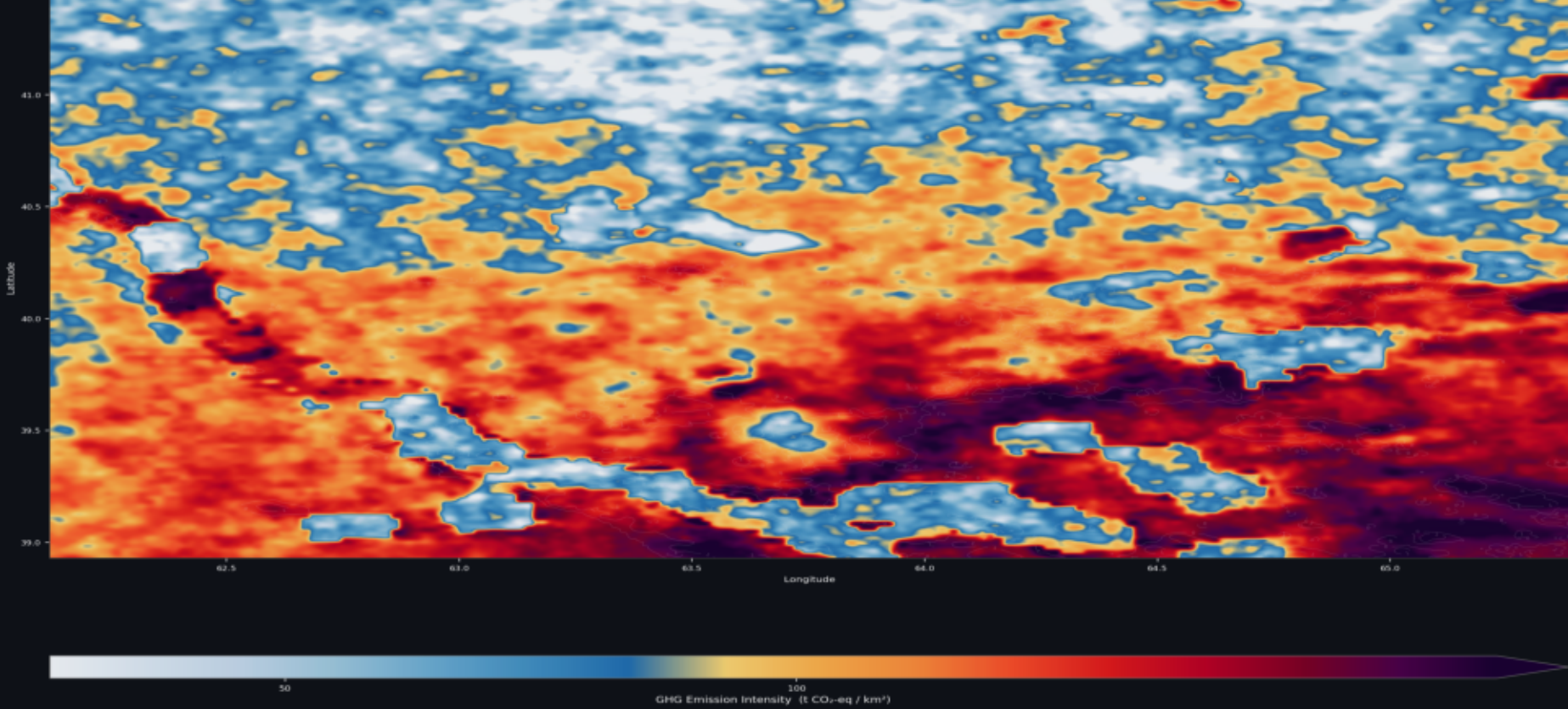


v2 — S5P + LULC + NL

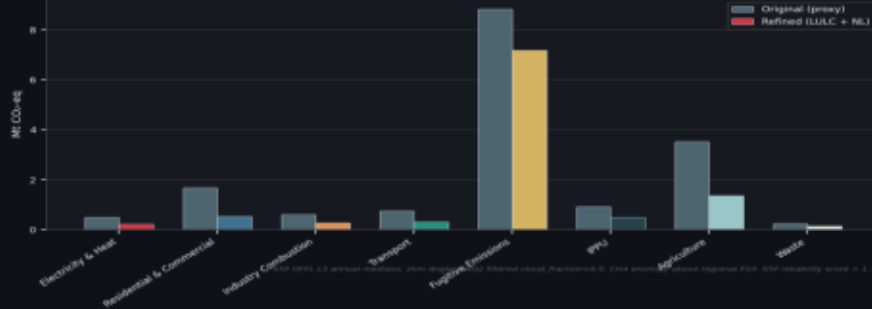
GHG Emissions — Bukhara Region (2022) — Sentinel-5P Refined

S5P weight: 32% of nominal (LULC-adjusted) | Spatial constraint: S5P NO2/SO2/CO/CH4 + ESRI 10m LULC + VIIRS NL | Total: BTR3 Uzbekistan 2024
Bukhara Region | Refined total: 10.47 Mt CO₂-eq (excl. LULUCF) | S5P-constrained spatial disaggregation

S5P reliability: LOW (32%)



Original vs Refined Sector Emissions

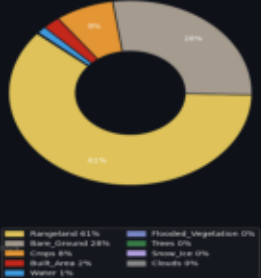


Source: ESRI Global LULC v2020 10m, NOAA VIIRS DMSP Monthly, BTR3 Uzbekistan to UNEPCE (2024), Proxy: Built+NL, Industrial, Crops (aggregations), Trees (LULUCF)

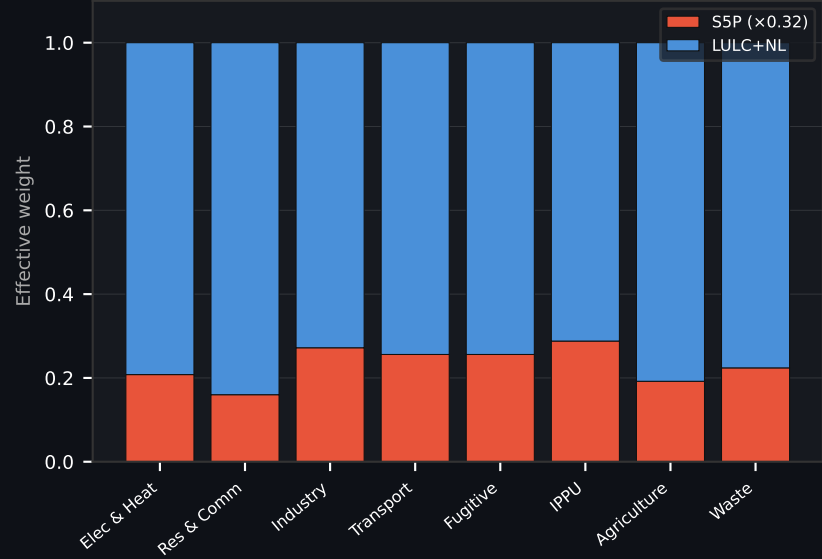
ESRI Land Cover Classes



Land Cover Composition



v2 S5P Blend Weights per Sector



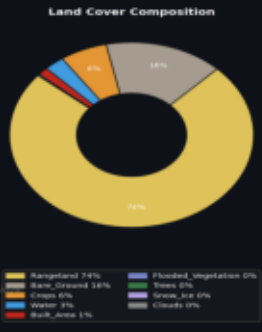
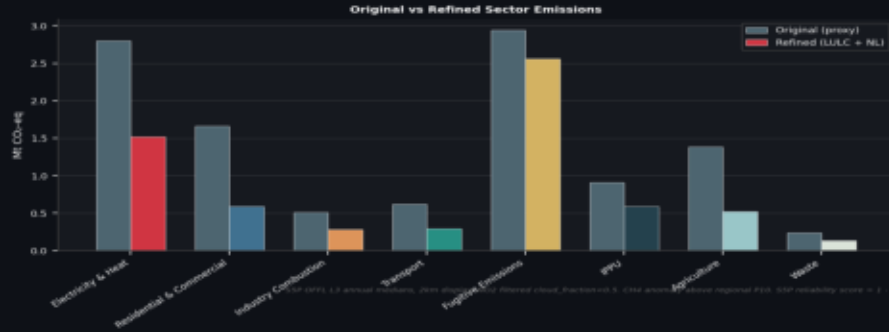
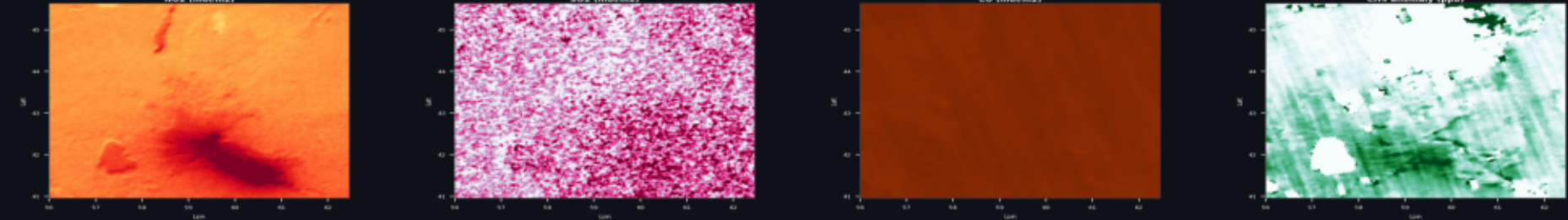
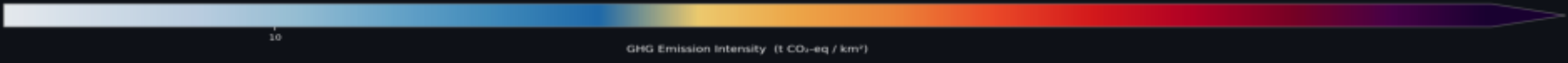
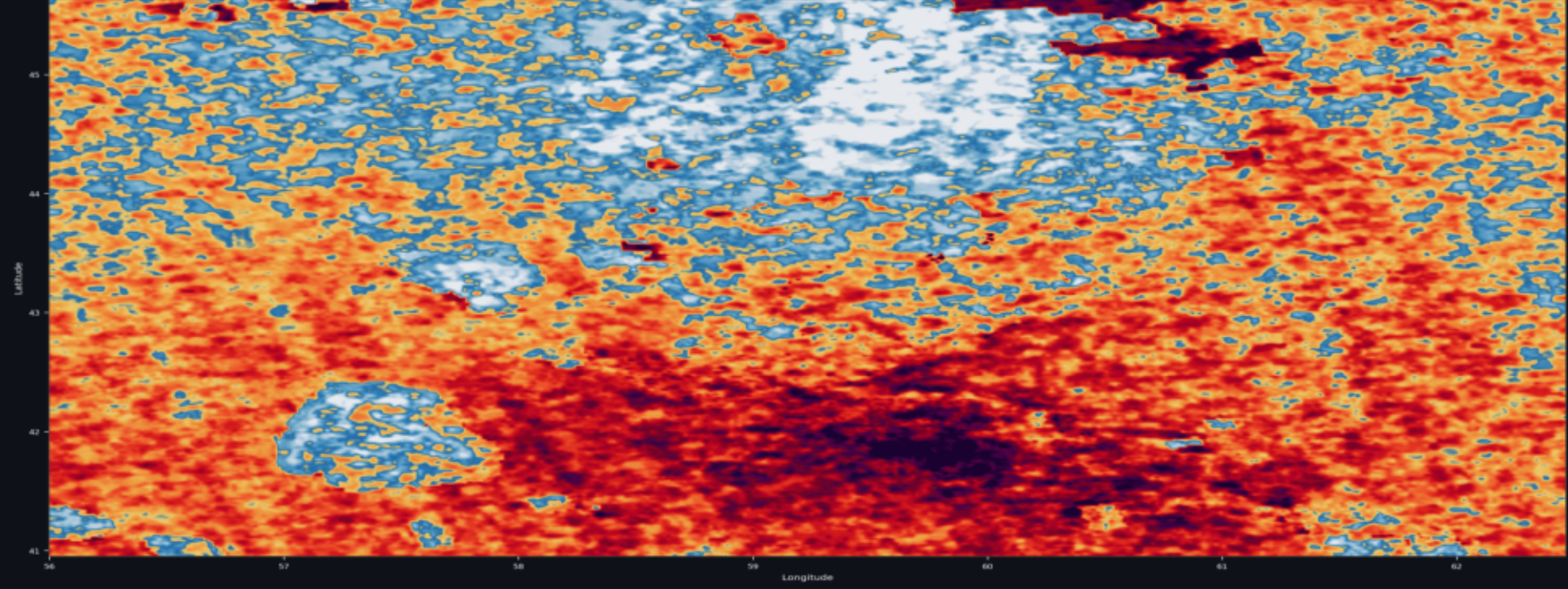
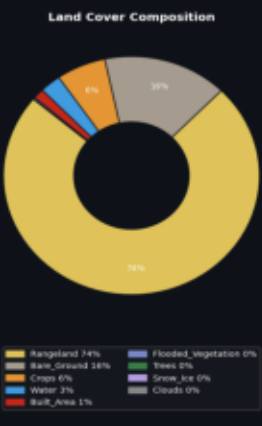
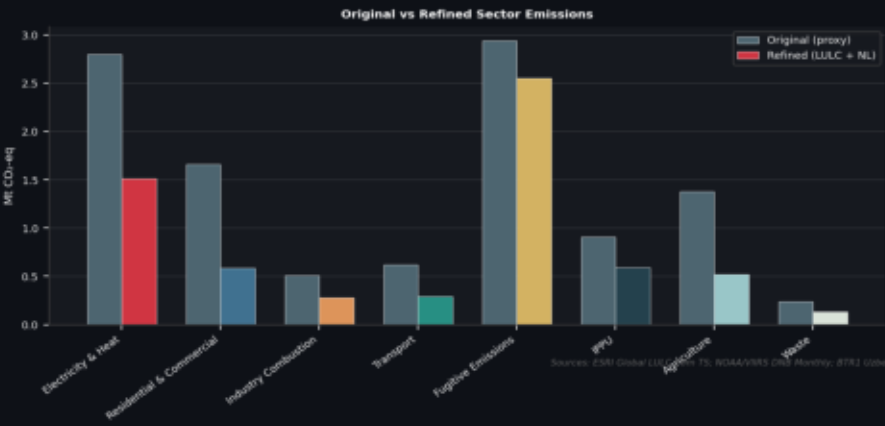
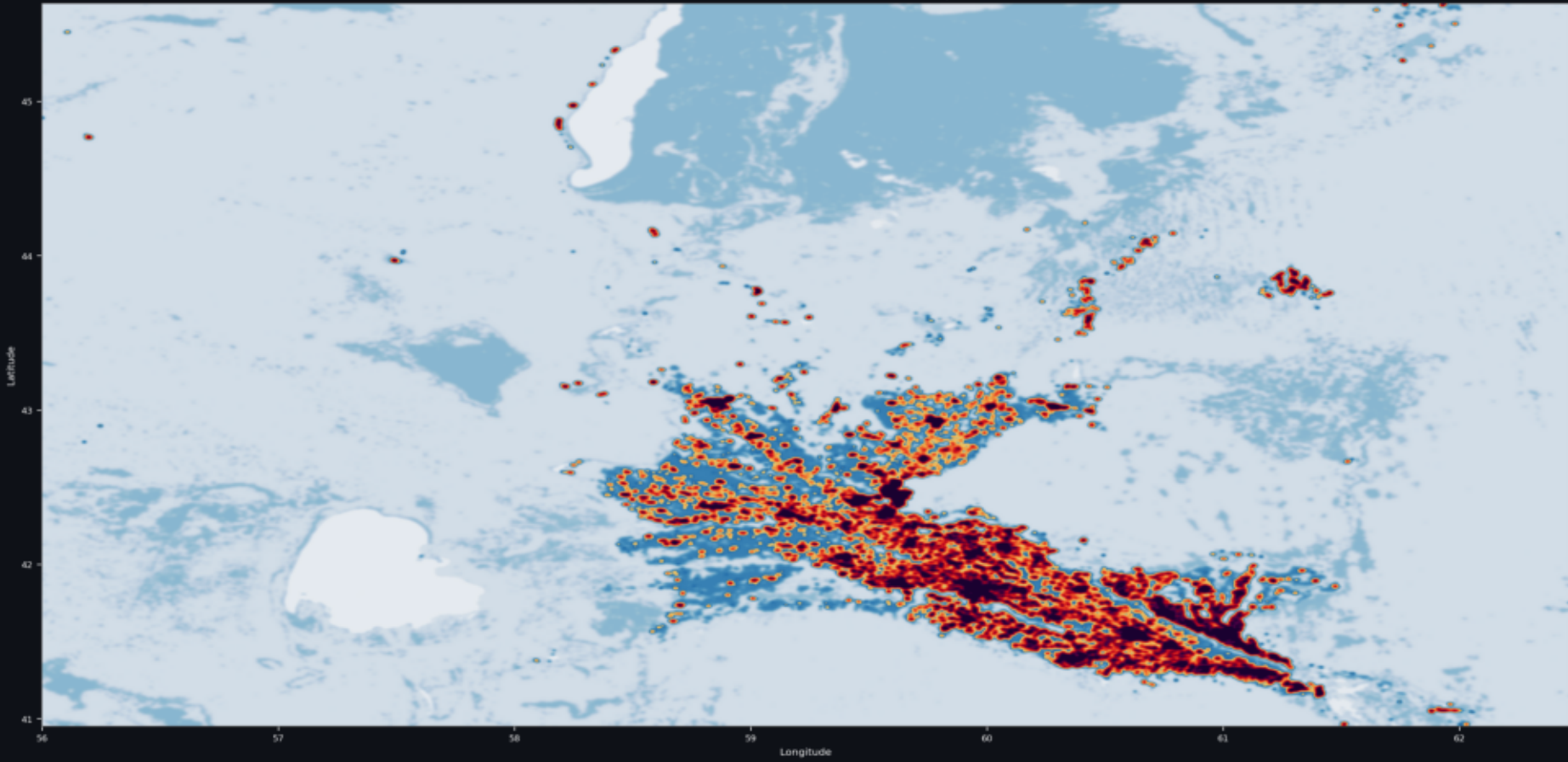
S5P Reliability:
LOW (32%)

Approach:
LULC/NL dominant

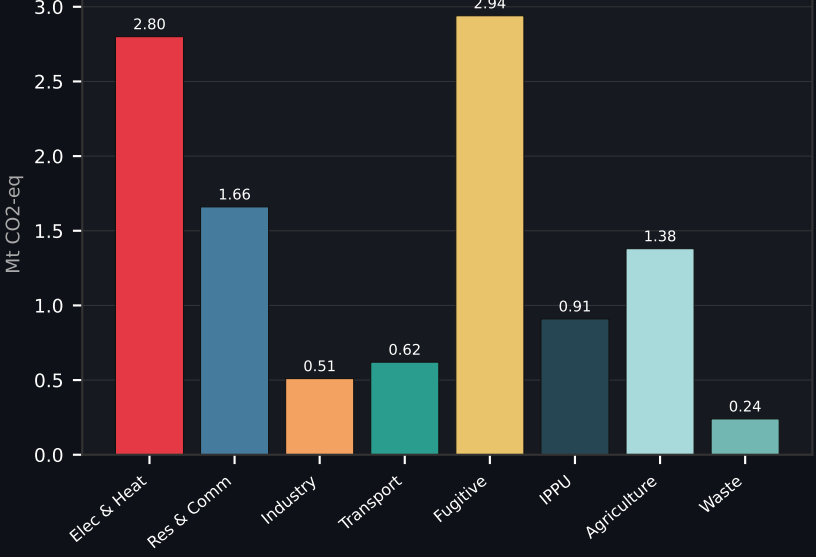
Total emissions:
17.02 Mt CO₂-eq

Dominant sector:
Fugitive Emissions

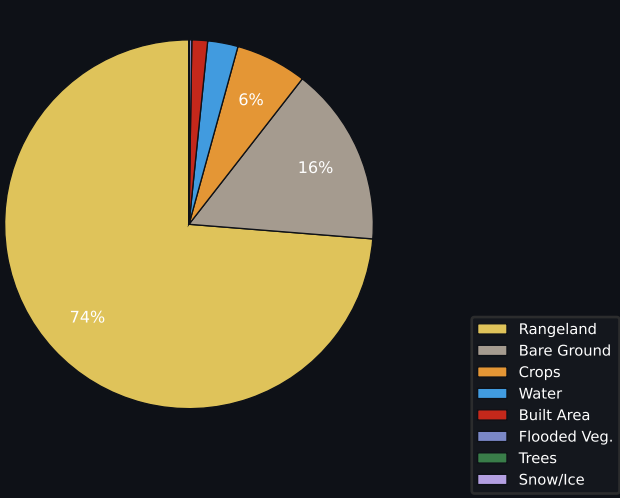
Dominant LULC:
Rangeland



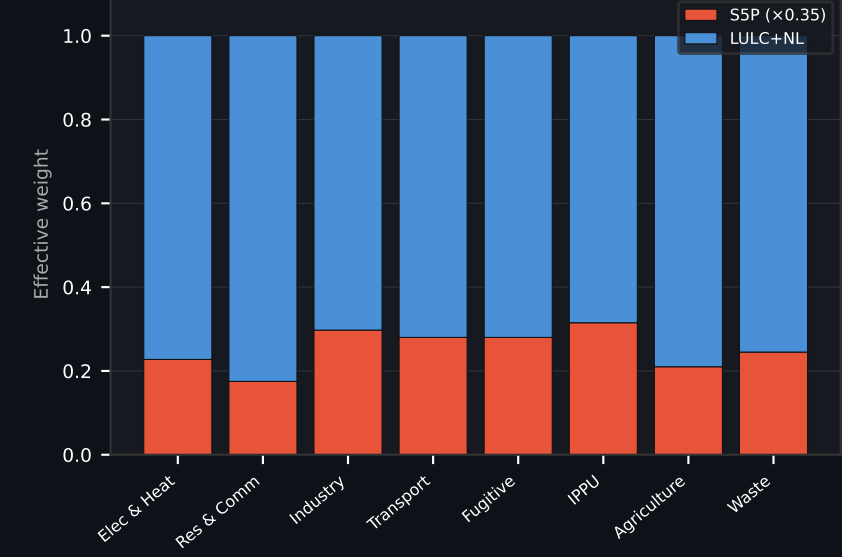
Sector Emissions (Mt CO₂-eq)



LULC Composition (%)



v2 S5P Blend Weights per Sector



S5P Reliability:

LOW (35%)

Approach:

LULC/NL dominant

Total emissions:

11.06 Mt CO₂-eq

Dominant sector:

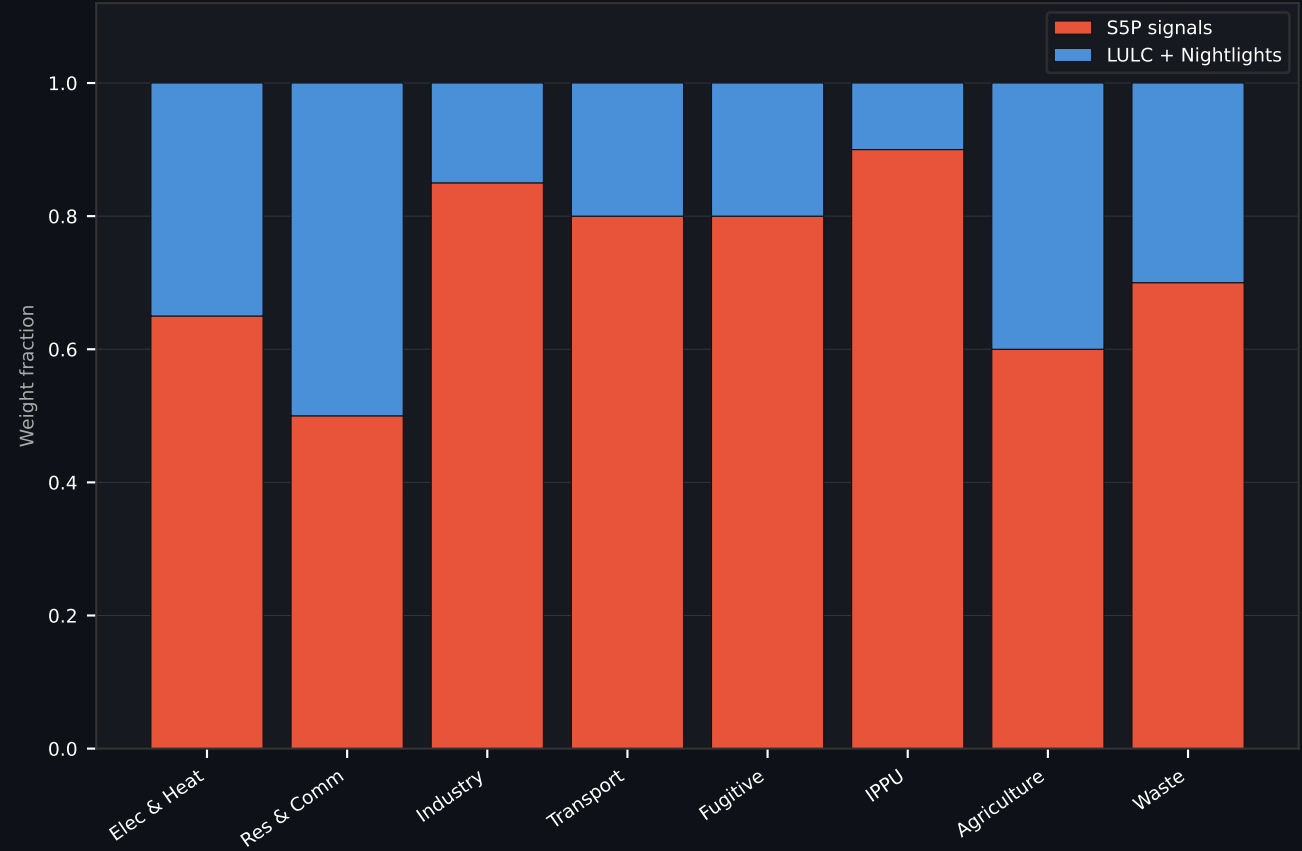
Fugitive Emissions

Dominant LULC:

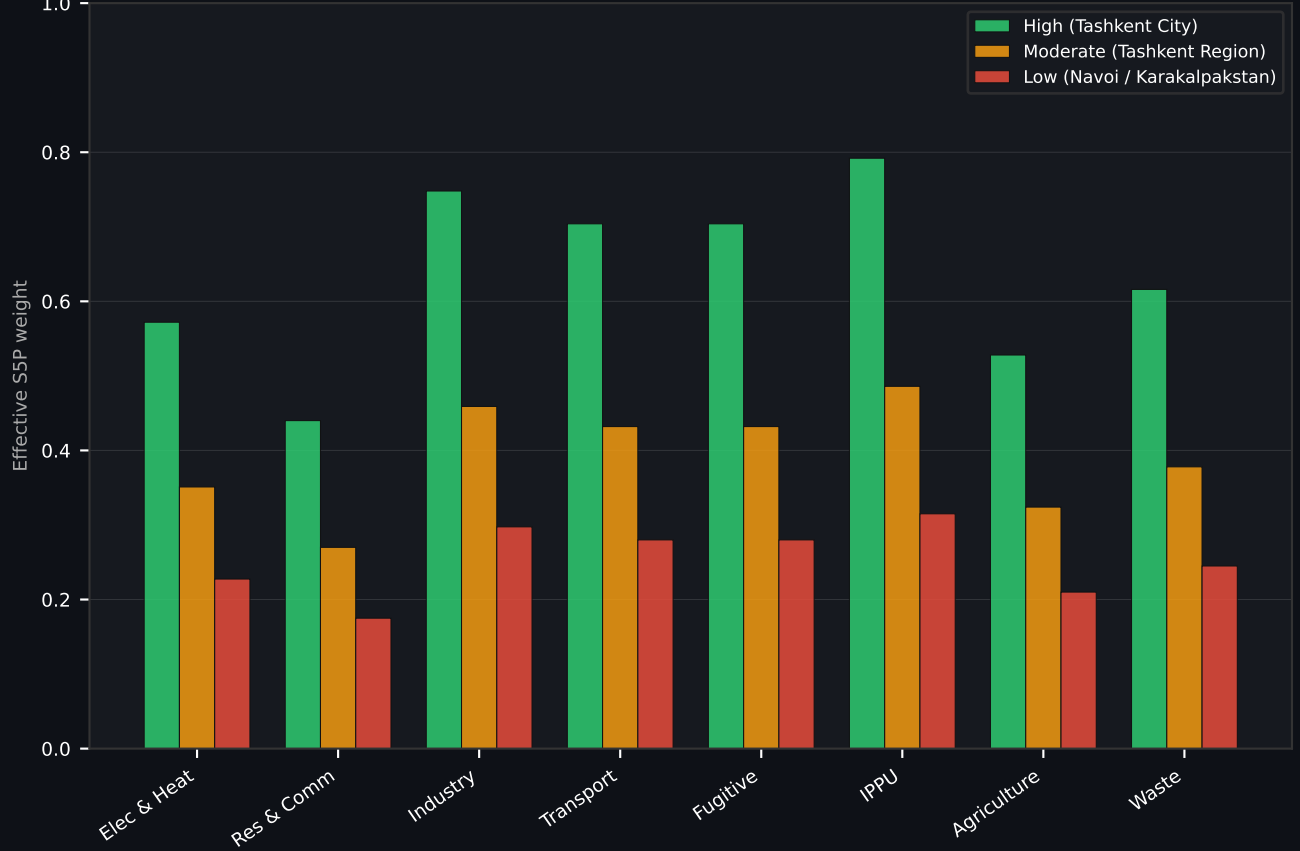
Rangeland

Methodology Analytics — Signal Weights & Regional Variation

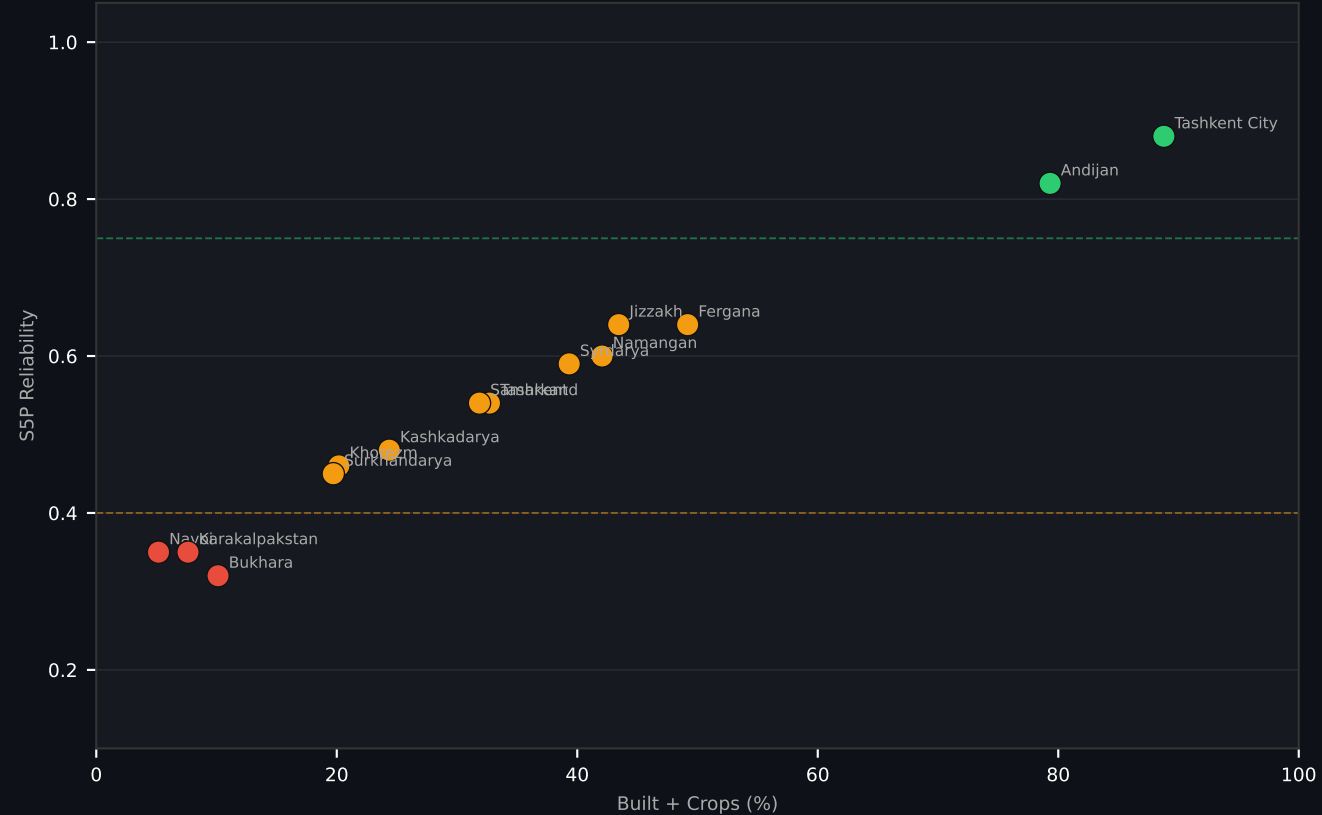
A — Nominal S5P vs LULC/NL weight per sector



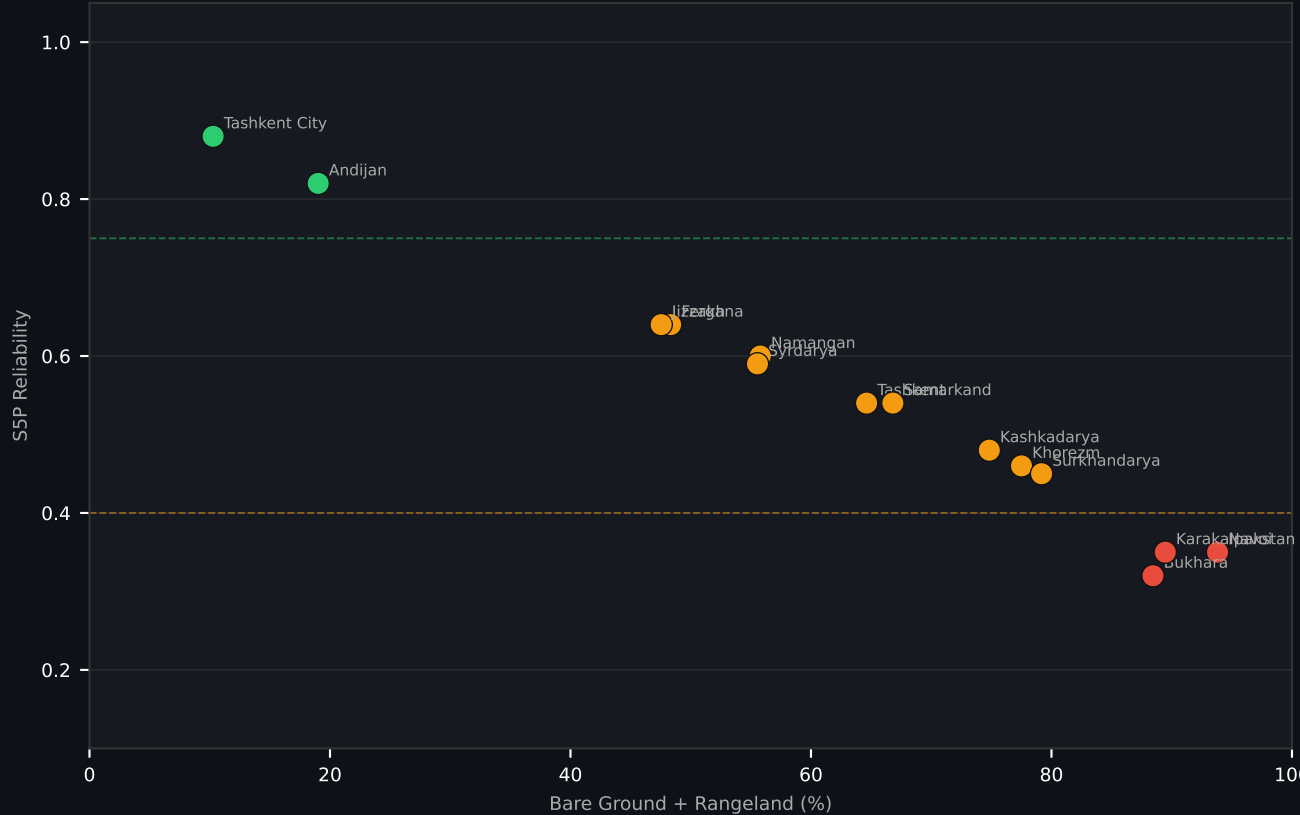
B — Effective S5P weight at different reliability levels



C — S5P Reliability vs Anthropogenic LULC fraction



D — S5P Reliability vs Arid LULC fraction



Key Findings & Recommendations

HIGH reliability — S5P as primary constraint

Regions: Tashkent City, Andijan Region
NO2 and SO2 column densities reflect genuine combustion activity. Traffic corridors, power plants, and industrial clusters are clearly resolved in the S5P field. The v2 spatial patterns are physically meaningful and preferred for hotspot identification and policy targeting.

MODERATE reliability — blended S5P + LULC/NL

Regions: Fergana Region, Jizzakh Region, Namangan Region, Syrdarya Region, Tashkent Region, Samarkand Region, Kashkadarya Region, Khorezm Region, Surkhandarya Region
S5P contributes at 40-74% of nominal weight after reliability scaling. Mixed signals from semi-arid rangeland, mountain terrain, or large rural areas reduce retrieval quality. Results are indicative; local inventory data should be used for validation where available.

LOW reliability — LULC + NL dominant (v1 preferred)

Regions: Navoi Region, Bukhara Region, Rep. Karakalpakstan
Desert surfaces (Kyzylkum, Karakum, Aral Sea basin) severely degrade TROPOMI UV/VIS retrievals. AER_AI is dominated by natural mineral dust. S5P weight is clamped to 15% of nominal. v1 (LULC+NL) gives a cleaner spatial pattern and should be the primary reference for these regions.

Application Guidance

Application	Method	Notes
* Urban hotspot identification	v2 (S5P NO2/SO2)	HIGH/MOD regions
Industrial emission cluster mapping	v2 (SO2 primary)	HIGH/MOD regions
* Traffic / transport corridor mapping	v2 (NO2 + CO)	HIGH/MOD regions
Fugitive / gas-field emission mapping	v2 (CH4 anomaly)	All regions
* Agricultural emission distribution	v2 (CH4 + crops)	All regions
Desert / arid region spatial pattern	v1 (LULC+NL)	LOW reliability
* Sub-city spatial detail (<1 km)	v1 (200m LULC)	S5P footprint ~5.5 km
Sector total allocation (budgets)	Either (identical)	Totals preserved in both

Data Sources

- Emission inventories: Uzbekistan First Biennial Transparency Report (BTR1), 2024
- Land cover: ESRI Global LULC 10m Time Series (projects/sat-io/open-datasets/landcover/ESRI_Global-LULC_10m_TS)
- Nightlights: NOAA/VIIRS DNB Monthly V1 VCMCFG, annual median 2022
- S5P NO2: COPERNICUS/S5P/OFFL/L3_NO2 — tropospheric column, cloud_fraction<0.5
- S5P SO2/CO/AER: COPERNICUS/S5P/OFFL/L3_SO2, L3_CO, L3_AER_AI
- S5P CH4: COPERNICUS/S5P/OFFL/L3_CH4 — bias-corrected, anomaly above regional P10
- Admin boundaries: FAO GAUL 2015 Level-1